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Enhancing few-shot continual relation extraction via pairwise augmentation and guided sharpness-aware optimization ⋆

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ABSTRACT

Few-shot Continual Relation Extraction (FCRE) remains a challenging problem due to the dual issues of limited annotated data and catastrophic forgetting. To address these challenges, we propose PRAGAS, a unified framework that integrates Pairwise Augmentation (PairAug) and Guided Sharpness-Aware Minimization (GuidedSAM) with Salient Distillation. PairAug enriches the training distribution by generating diverse, relation-preserving synthetic samples without relying on external models, while GuidedSAM adaptively adjusts optimization sharpness based on relational knowledge deviation, promoting stable and generalizable learning across tasks. Extensive experiments on FewRel and TACRED demonstrate that PRAGAS consistently surpasses prior state-of-the-art approaches by a remarkable margin in average accuracy while significantly mitigating forgetting. Comprehensive ablation studies further validate the complementary contributions of PairAug and GuidedSAM, the effectiveness of the proposed loss components, and the robustness of key hyperparameters. Moreover, we extend PRAGAS to Continual Event Detection (CED), where it achieves consistent improvements, underscoring its versatility as a general continual information extraction framework. All resources are publicly available at: https://github.com/datht4889/PRAGAS

1. Introduction

Relation Extraction (RE) is a fundamental task in Information Extraction that identifies semantic relationships between entities in text. Conventional RE methods typically depend on large amounts of annotated data and operate under the assumption of a fixed set of predefined relation types [1–3]. In practical scenarios, domains such as scientific research [4], medicine [5], and law [6] are evolving rapidly, resulting in a continual increase in the variety of relations [7,8]. Therefore, RE systems must demonstrate adaptability to effectively manage these dynamic developments. Moreover, a persistent challenge in RE model development lies in the limited availability of annotated data for newly emerging relations. These challenges have motivated the development of the Few-shot Continual Relation Extraction (FCRE) paradigm [9,10], which represents a difficult yet practically relevant research setting that has recently gained considerable attention. In this framework, a model is expected to incrementally learn novel relations from a limited

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number of examples while maintaining knowledge acquired from earlier tasks. Nevertheless, given the continual learning nature and scarcity of training data, FCRE models tend to exhibit bias toward the current task, thereby encountering issues of *overfitting* [11] and *catastrophic forgetting* [8,12–17].

To address these challenges, recent studies have adopted strategies inspired by traditional continual learning, with a predominant focus on memory-based approaches [7,9,10,18–21]. These methods generally retain a limited number of representative samples from previous tasks in a *replay buffer*, which are presented during the training of subsequent tasks, allowing the model to directly revisit prior knowledge while learning new relations. Although this approach has demonstrated effectiveness in numerous FCRE frameworks [9,10,22–25], it still exhibits several limitations. ***First***, in FCRE settings, the availability of samples for each relation category is extremely limited. As a result, models are typically fine-tuned using only a single instance per old class stored in the memory buffer, along with a small number of examples from newly introduced classes. This severe data scarcity not only restricts the model's ability to generalize but also increases the risk of overfitting and catastrophic forgetting. ***Second***, data scarcity and overfitting also often drive the optimization process toward sharp minima in the loss landscape [26,27]. Such sharp minima are known to hinder model generalization and exacerbate the forgetting phenomenon in continual learning. While this issue has been extensively discussed in the broader continual learning literature [14,28], it remains largely unexplored in the context of FCRE. To overcome these limitations, we propose PRAGAS, a framework that integrates **P**air**r**wise **A**ugmentation and **G**uided Sharpness-**A**ware Minimization with **S**alient Distillation, aiming to alleviate data scarcity and mitigate the sharp minima landscape problem in FCRE and even related Information Extraction (IE) tasks such as continual event detection (CED).

Specifically, to mitigate the impact of data scarcity, we propose a novel **Pairwise Augmentation** (PairAug) technique that enriches the training distribution by generating diverse, relation-preserving synthetic examples from the limited available data. Several existing approaches also rely on data augmentation [22,23], but they typically depend on costly external language models (e.g., ChatGPT) and inherit the limited diversity of the memory buffer. Our method, in contrast, operates entirely on the available datasets and is paired with specialized objective functions that act as a compositional, context-mixing regularizer on the relational representation. Together, these innovations substantially improve model performance, particularly by reducing overfitting in the challenging FCRE setting.

To further stabilize the learning process, we aim to identify flat minima in the loss landscape, a property shown to alleviate catastrophic forgetting [14,28]. While Sharpness-Aware Minimization (SAM) [29,30] is a natural candidate and complements our PairAug strategy, naively applying SAM in continual learning often yields suboptimal performance under the dynamic, non-stationary task stream (Fig. 3). We therefore introduce **GuidedSAM**, a trust-region-inspired optimizer [31] that dynamically adjusts the SAM perturbation radius via a **Salient Distillation** score measuring relational knowledge deviation between the previous and current model. This adaptive mechanism guides the model toward flatter, more generalizable minima while preserving previously learned knowledge, effectively mitigating catastrophic forgetting in the FCRE setting.

In summary, the key contributions of this work are as follows:

① We propose a novel data augmentation technique, Pairwise Augmentation, to enrich the limited datasets inherent in the FCRE scenario. This method exposes the model to diverse relational contexts and imposes explicit regularization on the relational representation through a dual-objective loss, substantially enhancing its capacity to reduce overfitting, alleviate forgetting, and improve overall performance.

② We incorporate Sharpness-Aware Minimization (SAM) and its variants to identify flatter minima in the loss landscape, enhancing model stability and generalization in the FCRE setting while mitigating catastrophic forgetting.

③ We introduce GuidedSAM, a trust-region-inspired optimizer that adaptively adjusts the perturbation radius based on a Salient Distillation score measuring relational knowledge deviation between models. This dynamic mechanism effectively balances stability and plasticity, preserving prior knowledge and preventing overfitting to new relations.

④ We conduct extensive experiments on two benchmark FCRE datasets, FewRel and TACRED, demonstrating that our proposed method consistently outperforms prior state-of-the-art approaches by a substantial margin of 3–4% in average accuracy. Our analysis on the proposed components further reveals their individual and complementary contributions to mitigating forgetting and improving generalization. Moreover, we extend our framework to Continual Event Detection (CED), showcasing its strong adaptability and generalization capability across continual information extraction tasks.

Structure. The remainder of this paper is organized as follows. Section 2 reviews the related work in FCRE. Section 3 introduces the necessary background and problem formulation for few-shot continual relation extraction (FCRE). Section 4 presents the details of our proposed PRAGAS framework. Section 5 describes the experimental setup, datasets, baselines, and evaluation results, followed by in-depth analyses and ablation studies. Finally, Section 6 concludes the paper and discusses potential directions for future research. The Appendix provides additional results demonstrating the effectiveness of our approach on the Continual Event Detection (CED) task.

Key extensions over prior works. A preliminary version of this work appeared as Anh et al. [32] and introduced the Pairwise Augmentation (PairAug) component. The present manuscript substantially extends that conference version along the following axes. **(i) A new optimization mechanism, GuidedSAM with Salient Distillation** (Section 4.2), accompanied by an **empirical study of SAM under data scarcity in FCRE** (Section 4.3) that motivates its design and reveals counter-intuitive behaviors of vanilla SAM

in this regime; neither component is present in the prior work. **(ii) Generalization to advanced LLM-based backbones** (BGE [33] and three LLM2Vec [34] variants based on Mistral, Llama2, and Llama3), demonstrating encoder-agnostic effectiveness beyond the BERT-only setting evaluated in the prior work. **(iii) Cross-task generalization to Continual Event Detection** (Appendix), confirming applicability of the framework beyond relation extraction. The PairAug component, inherited from the prior work, is also formalized more precisely in Section 4.1.1 to match its implementation.

2. Related work

2.1. Continual learning

Continual Learning (CL), also known as lifelong learning, addresses the challenge of training models on a sequence of tasks without forgetting previously acquired knowledge [12]. The primary obstacle in CL is *catastrophic forgetting* [35], where a model's performance on old tasks degrades significantly after learning new ones. Broadly, CL methods can be categorized into three families:

1. *Regularization-based methods* introduce a penalty term into the loss function to constrain updates to parameters deemed important for previous tasks. Techniques like Elastic Weight Consolidation (EWC) [36] and Synaptic Intelligence (SI) [37] estimate parameter importance and penalize large changes to critical weights.

2. *Architecture-based methods* adapt the model's structure to accommodate new knowledge, for instance, by allocating new subnetworks for new tasks [38] or by learning task-specific masks over a fixed, larger architecture [39]. While effective, these methods often lead to an increase in model size over time.

3. *Memory-based (or replay) methods* store a small subset of data from past tasks in an episodic memory buffer [40,41]. During training on a new task, samples from this buffer are replayed alongside the new data, which helps to alleviate forgetting. These methods have proven particularly effective and popular in Natural Language Processing (NLP) tasks [42,43].

Our work belongs to the family of memory-based approaches, yet it distinguishes itself by emphasizing the maximization of each replayed sample's utility through the integration of advanced augmentation and adaptive optimization strategies.

2.2. Relation extraction

Traditional relation extraction. Relation Extraction (RE) is a fundamental task in Natural Language Processing (NLP) that aims to identify entities and the semantic relationships between them within a given text [44,45]. Each relation is typically represented as a triplet $\langle e_h$, relationship, $e_t\rangle$, denoting a pair of entities and the relation connecting them. For example, in the sentence "Kamala Harris and Donald Trump were political opponents in the US presidential election", an RE model should recognize the "political opponents" relation between the entities "Kamala Harris" and "Donald Trump". Traditional RE systems, however, operate under a closed-world assumption, being trained once on a fixed dataset with a predefined set of relation types, which limits their adaptability to evolving or unseen relations in real-world scenarios.

Formally, given a natural language text x, the objective of the RE task is to predict a set of triplets $\langle e_h, r, e_t\rangle$, where e_h and e_t denote the head and tail entities, and r represents the relation connecting them. The entities e_h and e_t may correspond to words, phrases, or other syntactic units in the text, while the relation $r \in R$ is selected from a predefined set of relation types that describe the semantic association between e_h and e_t.

Few-shot continual relation extraction (FCRE). While traditional CL often assumes ample data for each task, many real-world scenarios present new information in a data-scarce manner. This has led to the development of Few-shot continual relation extraction (FCRE). FCRE sits at the intersection of few-shot learning and continual (lifelong) learning in NLP. FCRE models are trained sequentially on a stream of tasks, each introducing a small set of new relation types, and must quickly learn these relations from very few examples while retaining knowledge of previously learned relations [9,10]. This setting is motivated by real-world applications where relation schemas evolve over time and only a handful of labeled instances are available for new relations (Fig. 1). In particular, models must counteract catastrophic forgetting, which tends to lose performance on old relations when learning new ones, while also coping with the extreme data sparsity of few-shot tasks. Standard memory-based continual-learning approaches (e.g., rehearsal buffers or stored prototypes) tend to overfit when only a few examples are available, yielding weak recall of old relations and confusion among similar relation classes [10,18,25,46]. For instance, [18] proposed a serial replay mechanism and knowledge distillation, while [10] focused on maintaining consistent relation representations across tasks. More recently, rehearsal-free prompt-based methods have emerged as an alternative that avoids storing exemplars altogether [8,47]. Given the extreme data scarcity in FCRE, several works have explored data augmentation to enrich the training distribution. A recent trend involves using large language models (LLMs) as external knowledge sources. For example, Ma et al. [23] and Tran et al. [22] leverage models like ChatGPT to paraphrase or generate new training instances from the few samples stored in the memory buffer. While these methods can be effective, they introduce a significant dependency on costly, large-scale external models. Furthermore, the quality and diversity of the generated data are often constrained by the limited variety of the seed samples in the buffer. In contrast, our proposed PairAug tackles the same data scarcity challenge but operates exclusively on the available data, eliminating the need for external resources and ensuring more consistent and contextually relevant augmentation.

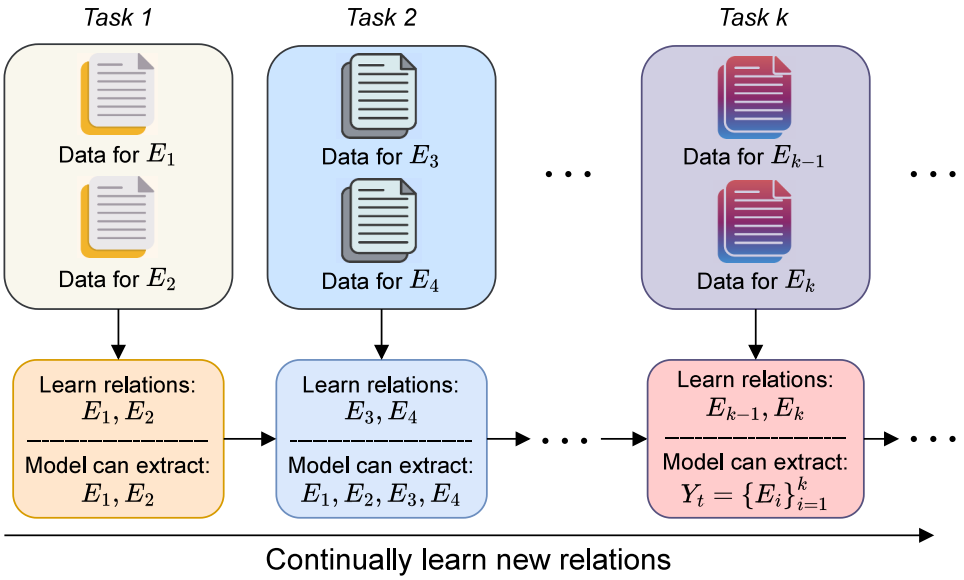


Fig. 1. Few-shot continual relation extraction (FCRE): a model incrementally learns new relation types from limited data as corresponding samples become available. At each task k, the model preserves its ability to recognize all previously encountered relation types while adapting to newly introduced ones.

3. Background

3.1. Problem formulation

In the Few-Shot Continual Relation Extraction (FCRE) setting, a model incrementally learns relational knowledge through a sequence of tasks. For each task $\mathcal{T}^t$, the model is trained on a dataset $\mathcal{D}^t = \{(x_i^t, y_i^t)\}_{i=1}^{N\times K}$, which comprises N newly introduced relations belonging to the relation set R^t, each containing K training instances. Specifically, each instance (x_i^t, y_i^t) consists of a sentence x_i^t containing an entity pair (e_h, e_t) and its corresponding relation label $y_i^t \in R^t$. This setup follows the conventional "N-way-K-shot" learning paradigm.

After the completion of task $\mathcal{T}^t$, the corresponding dataset $\mathcal{D}^t$ is no longer accessible in subsequent training phases. The model is then evaluated on the test sets of all tasks encountered so far, requiring it to correctly identify relations within the expanded relation set $\widetilde{R}^t = \bigcup_{i=1}^{t} R^i$.

Following prior work [9,18,42], we maintain a memory buffer $\mathcal{M} = \{\mathcal{M}^1, \mathcal{M}^2, \ldots, \mathcal{M}^t\}$ that stores a small number of representative samples from previously learned tasks. Specifically, for each relation, m representative examples are preserved in memory. During the training of task $\mathcal{T}^t$, the model is allowed to access all retained samples from previous tasks, denoted as $\widehat{\mathcal{M}}^{t-1} = \bigcup_{i=1}^{t-1} \mathcal{M}^i$. For brevity, we define the complete training data used in task t as $\mathcal{D}^t_{\text{train}} = \mathcal{D}^t \cup \widehat{\mathcal{M}}^{t-1}$. In the few-shot configuration, the memory size is highly constrained, and following prior work Qin and Joty [9], we store only one exemplar per relation, i.e., $m = 1$. This is moreover the hardest non-trivial memory budget, i.e., the smallest replay footprint short of using no buffer at all ($m = 0$), so it stresses catastrophic forgetting and data scarcity in their sharpest form at minimal storage overhead. We do not claim $m = 1$ is the only realistic setting; a larger budget ($m > 1$) is feasible and would likely benefit all methods, so $m = 1$ serves as a conservative lower bound rather than the sole regime of interest.

3.2. Input formulation and representation

In Relation Extraction (RE), foundational deep learning methods [48,49] often leverage pre-trained language models (PLMs) such as BERT [50] to encode input sequences. A key challenge in RE lies in designing effective input formats to generate high-quality feature representations suitable for relation classification. Traditional approaches typically append the `[CLS]` token to the input sequence x, relying on the embedding of this token for classification. Alternatively, some methods introduce special markers around the head and tail entities, using their respective embeddings—often concatenated—as input to the relation classification layer [7,51].

In this work, we adopt the input formatting strategy proposed by [23]. Specifically, we insert a dedicated `[MASK]` token to represent the target relation between the head entity (e_h) and the tail entity (e_t), and combine it with the original sentence x and the two entities. Additionally, to reduce dependence on manually designed templates, we incorporate several learnable tokens into the input.

The resulting input template is defined as follows:

$$\mathcal{T}(x) = x\Big[v_{0:n_0-1}\Big]e_h\Big[v_{n_0:n_1-1}\Big]\texttt{[MASK]} \Big[v_{n_1:n_2-1}\Big]e_t\Big[v_{n_2:n_3-1}\Big]. \tag{1}$$

where each $[v_i]$ denotes a learnable continuous token, and the indices n_i specify the length of token spans. In our implementation, we use a special [UNUSED] token to serve as the learnable placeholder for $[v]$.

The constructed input $\mathcal{T}(x)$ is then passed through a PLM, which encodes it into a sequence of contextualized embeddings. From this sequence, we extract the hidden representation $\boldsymbol{z}_x$ corresponding to the position of the [MASK] token:

$$\boldsymbol{z}_x = f_{\mathcal{M}}\big(\mathcal{T}(x)\big)[\text{index}(\texttt{[MASK]})], \tag{2}$$

where $f_{\mathcal{M}}(X)$ denotes the output of PLM $\mathcal{M}$ for input X. This latent embedding $\boldsymbol{z}_x$ is then used for both contrastive learning and relation prediction.

3.3. Constrastive learning-based FCRE frameworks

Contrastive learning has emerged as a foundational paradigm in FCRE due to its ability to structure the representation space effectively under limited supervision. By encouraging instances of the same relation to form compact clusters while separating those of different relations, contrastive objectives help maintain discriminative embeddings across tasks. Recent FCRE frameworks integrate contrastive learning either as a primary training objective or as a regularization strategy to enhance both stability and generalization [22–25]. Our proposed PRAGAS is orthogonal to and complementary with existing contrastive learning-based frameworks, making it universally applicable and capable of further enhancing their robustness and performance in the FCRE setting. In our experiments, we integrate PRAGAS into both the Contrastive Prompt Learning (CPL) [23] and SIRUS [25] frameworks to demonstrate its generality and effectiveness. Both these method formulate and represent the input as detailed in Section 3.2.

Contrastive prompt learning framework (CPL). Ma et al. [23] proposed techniques including prompt design, representation learning loss, and memory augmentation strategy to address the challenges of catastrophic forgetting and overfitting in FCRE. CPL employs ChatGPT to generate diverse synthetic samples through carefully designed prompts, aiming to alleviate overfitting in low-resource FCRE scenarios. The augmented data are then combined with the original samples to construct an enriched training set used for memory replay and continual learning.

For representation learning, CPL leverages a margin-based contrastive learning (MCL) objective function, where the loss for each sample x_i is formulated as follows:

$$L_{\text{CPL}}(i) = -\sum_{p\in P(i)} \log \frac{\exp(\alpha_{i,p}\cdot s_{i,p}/\tau)}{\sum_{a\in I}\exp(\alpha_{i,a}\cdot s_{i,a}/\tau)} \tag{3}$$

where $P(i) = \{p \in I : y_p = y_i\}$ is the index set of positive samples w.r.t sample x_i, $I = \{1, 2, \dots, n\}$ is index set of all n training samples; $s_{i,p} = \frac{z_i \cdot z_p}{||z_i||\cdot||z_p||}$ is the similarity between the representations z_i and z_p (Eq. (2)) of samples x_i and x_p, respectively; $\alpha_{i,p}$ is a relaxation factor, and τ is a temperature parameter.

Similar relation clusters (SIRUS). Le et al. [25] aim to address the challenge of highly similar relation classes by leveraging dynamic clustering based on relation descriptions. Specifically, they propose the Clustering Relations via Label Description technique, which applies the Agglomerative Clustering algorithm [52] to the description embeddings. These embeddings are dynamically updated with each training batch to identify informative hard negatives and enhance inter-class distinction. After identifying analogous relations, SIRUS incorporates several contrastive-based loss functions to further separate sample representations among similar classes:

- *Weighted Supervised Contrastive Loss* aims to pull samples from the same relation (positives) closer together while pushing samples from different relations (negatives) apart. Crucially, it assigns a higher weight to hard negatives (samples are in the same cluster but have different relation labels) to focus the model on distinguishing between similar relations.

$$\mathcal{L}_{\text{WSC}}(x) = -\sum_{p\in P(x)} \log \frac{f(\mathbf{z}_x, \mathbf{z}_p)}{\sum_{\bar{x}\in D\setminus\{x\}} w(x,\bar{x})\cdot f(\mathbf{z}_x, \mathbf{z}_{\bar{x}})} \tag{4}$$

where $f(\mathbf{z}_x, \mathbf{z}_y) = \exp\frac{\gamma(\mathbf{z}_x, \mathbf{z}_y)}{\tau}$ and

$$w(x,\bar{x}) = \begin{cases} 1 + \alpha\cdot\gamma(\mathbf{d}_x, \mathbf{d}_{\bar{x}}), & \text{if } c(r_x) = c(r_{\bar{x}}) \\ 1, & \text{otherwise} \end{cases} \tag{5}$$

Here, $\gamma(\cdot,\cdot)$ represents the Cosine Similarity function, τ is the temperature scaling parameter, α is a weighting factor based on the description similarity, and $c(\cdot)$ is the cluster assignment function. $P(x)$ and D refer to the sets of positive samples for sample x and the entire dataset, respectively.

- *Cluster-based Mutual Information Loss* works to maximize the mutual information between a sample's embedding ($\mathbf{z}_x$) and its corresponding label description embedding ($\mathbf{d}_x$). It pushes the sample's embedding away from the descriptions of negative relations, applying a similar weighting scheme as $\mathcal{L}_{\text{WSC}}$ to emphasize hard negatives found within the same cluster.

$$\mathcal{L}_{\text{CMI}}(x) = -\log \frac{h(\mathbf{z}_x, \mathbf{d}_x)}{Z(x)} \tag{6}$$

- *Double Triplet Loss for Intra- and Inter-Cluster Separation* $\mathcal{L}_{\text{DT}}$ enforces two distance constraints. First, it ensures a sample's embedding ($\mathbf{z}_x$) is closer to its own relation's description ($\mathbf{d}_x$) than to its cluster's centroid ($\mathbf{c}_x^+$). Second, it pushes the sample's embedding to be closer to its own cluster's centroid ($\mathbf{c}_x^+$) than to the centroid of the nearest negative cluster ($\mathbf{c}_x^-$).

$$\mathcal{L}_{\text{DT}}(x) = \max\left(0, D(\mathbf{z}_x, \mathbf{d}_x) - D(\mathbf{z}_x, \mathbf{c}_x^+) + m_1\right) + \max\left(0, D(\mathbf{z}_x, \mathbf{c}_x^+) - D(\mathbf{z}_x, \mathbf{c}_x^-) + m_2\right) \tag{7}$$

Finally, the total loss function is formulated by combining all three losses:

$$\mathcal{L}_{\text{SIRUS}} = \lambda_1 \mathcal{L}_{\text{WSC}} + \lambda_2 \mathcal{L}_{\text{CMI}} + \lambda_3 \mathcal{L}_{\text{DT}} \tag{8}$$

3.4. Sharpness-aware minimization (SAM) and variants

Sharpness-aware minimization (SAM). The generalization ability of deep neural networks has been shown to correlate strongly with the flatness of the loss minima they converge to [53]. Sharpness-Aware Minimization (SAM) [54] is an optimization technique designed to explicitly seek such flat regions, thereby improving both robustness and generalization. This characteristic is particularly desirable in continual learning (CL), where the data distribution shifts inherently from one task to the next [55]. Intuitively, a model that has converged to a flat minimum on a previous task is less likely to be perturbed into regions of high error when adapting to new tasks. Although the application of SAM to CL is still emerging, recent studies have started to investigate its effectiveness in mitigating forgetting and enhancing stability [31].

Let the overall loss function be $\mathcal{L}$, and consider a data batch B. SAM improves generalization by minimizing the worst-case loss within a neighborhood around the current model parameters, thereby steering training toward flatter minima [54]. Formally, the SAM objective is expressed as:

$$\min_{\theta} \max_{\|\epsilon\|_2 \le \rho} \mathcal{L}(\theta + \epsilon) \tag{9}$$

The perturbation ϵ is constrained to lie within an ℓ_2 Euclidean ball of radius ρ. In practice, the SAM minimax problem is optimized using an iterative two-step procedure for $t = 0, 1, 2, \ldots$, as follows:

$$\begin{cases} \epsilon_t = \rho \dfrac{\nabla \mathcal{L}_B(\boldsymbol{\theta}_t)}{\|\nabla \mathcal{L}_B(\boldsymbol{\theta}_t)\|_2} \\ \boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \alpha_t\left(\nabla \mathcal{L}_B(\boldsymbol{\theta}_t + \epsilon_t)\right) \end{cases} \tag{10}$$

where $\nabla \mathcal{L}_B$ denotes the gradient computed on the minibatch B, and α_t is the learning rate at step t.

Adaptive sharpness-aware minimization (ASAM) [29]. ASAM addresses a key limitation of SAM: sensitivity to parameter re-scaling. The authors state that neural networks with parameters θ and $A\theta$ (where A is a scaling operator that does not change the loss function) can have arbitrarily different sharpness values, despite having the same generalization gap. This scale-dependency weakens the correlation between sharpness and generalization.

To solve this, ASAM introduces "adaptive sharpness" , a scale-invariant measure that normalizes the search space for the perturbation. This results in the following minimax objective (shown for $p = 2$):

$$\min_{\theta} \max_{\|T_\theta^{-1}\epsilon\|_2 \le \rho} \mathcal{L}_B(\theta + \epsilon) + \frac{\eta}{2}\|\theta\|_2^2 \tag{11}$$

where T_θ is a normalization operator, often defined as a diagonal matrix of the absolute weight values, $T_\theta = \text{diag}(|\theta_1|, \ldots, |\theta_k|)$. This adaptive normalization leads to the following two-step update rule (for $p = 2$):

$$\begin{cases} \epsilon_t = \rho \dfrac{T_{\theta_t}^2 \nabla \mathcal{L}_B(\boldsymbol{\theta}_t)}{\|T_{\theta_t} \nabla \mathcal{L}_B(\boldsymbol{\theta}_t)\|_2} \\ \boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \alpha_t\left(\nabla \mathcal{L}_B(\boldsymbol{\theta}_t + \epsilon_t) + \eta\, \theta_t\right) \end{cases} \tag{12}$$

By making the perturbation region adaptive to the weight scale, ASAM achieves a stronger correlation with generalization and improves model performance.

Trust region aware minimization (TRAM) [31]. Expanding from ASAM, TRAM introduced an awareness of function curvature (i.e., the trust region) into sharpness-aware minimization. TRAM's core innovation is to replace the fixed neighborhood radius ρ in Eq. (12) with the trust region metric, denoted as d. This metric can be estimated using the KL divergence from a prior model distribution or the divergence from the current distribution under parametric input noise. By substituting ρ with d, TRAM constrains the maximization domain for the ascent step (i.e.,$\theta \rightarrow \theta + \epsilon$) to a parameter neighborhood that is explicitly constrained by low representation curvature

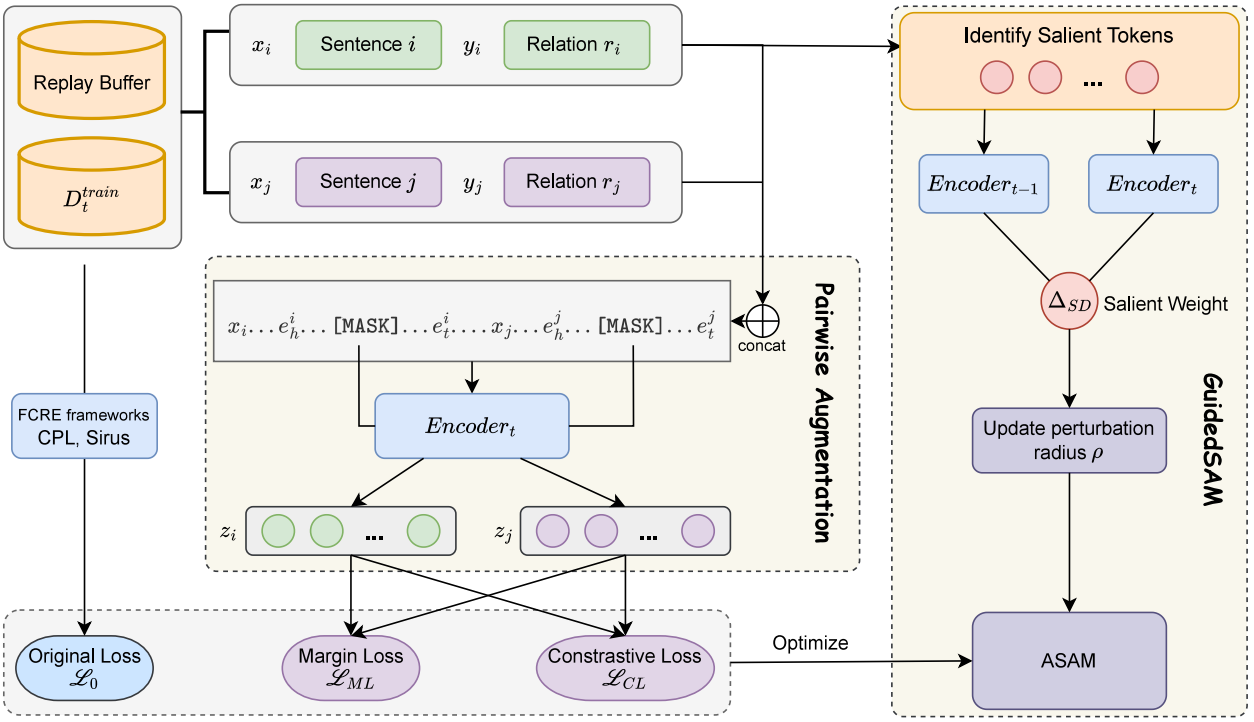


Fig. 2. Overview of the proposed PRAGAS framework. Pairwise augmentation generates augmented relational inputs from current and replay samples, optimized via $\mathcal{L}_0$, $\mathcal{L}_{ML}$, and $\mathcal{L}_{CL}$. GuidedSAM identifies salient tokens and uses the Salient Distillation score (Δ_{SD}) to adaptively adjust the perturbation radius ρ in ASAM, enhancing generalization and mitigating forgetting.

i.e., $\max_{\|\epsilon\|_2 \le d}$. This modification forces the optimizer to find minima that are not only flat in the parameter space (sharpness-awareness) but also smooth in the function space (curvature-awareness). This stands in contrast to SAM and ASAM, whose ρ-bounded neighborhoods are insensitive to the underlying function geometry.

The resulting TRAM objective, which pursues a minimum robust to curvature-aware perturbations, is expressed as:

$$\min_{\theta} \max_{\|\epsilon\|_2 \le d} \mathcal{L}_B(\theta + \epsilon) + \frac{\eta}{2}\|\theta\|_2^2 \tag{13}$$

TRAM solves this min-max problem in two stages, following the SAM/ASAM procedure. First, an ascent step computes the optimal perturbation ϵ_t that maximizes the loss within the trust region d. Second, a descent step updates the model parameters using the gradient evaluated at this new perturbed point. The complete update rule is:

$$\begin{cases} \epsilon_t = d\, \dfrac{\theta^2 \nabla \mathcal{L}_B(\boldsymbol{\theta}_t)}{\|\theta \nabla \mathcal{L}_B(\boldsymbol{\theta}_t)\|_2} \\ \boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \alpha_t \big(\nabla \mathcal{L}_B(\boldsymbol{\theta}_t + \epsilon_t) + \eta\, \theta_t\big) \end{cases} \tag{14}$$

4. Methodology

In this section, we introduce PRAGAS, a unified framework that enhances knowledge retention and generalization across sequential tasks. Our approach builds upon the memory-based paradigm but introduces novel mechanisms to maximize the utility of stored samples through efficient augmentation and guided optimization. The overall workflow of the proposed PRAGAS framework is illustrated in Fig. 2. Specifically, our method first applies **Pairwise Augmentation** (Section 4.1), where two sentences from the current training set and replay buffer are concatenated to create augmented relational inputs. The encoder extracts their representations, which are optimized jointly using the margin loss $\mathcal{L}_{ML}$ and contrastive loss $\mathcal{L}_{CL}$ alongside the base FCRE loss $\mathcal{L}_0$. In parallel, GuidedSAM (Section 4.2) component inspired by trust-region learning to identify salient tokens and compute the Salient Distillation score (Δ_{SD}) between the current and previous encoders to adaptively update the perturbation radius ρ in ASAM. Together, these components improve generalization, mitigate overfitting, and alleviate catastrophic forgetting in FCRE.

4.1. Mitigating data scarcity via pairwise augmentation

A core challenge in FCRE lies in the limited labeled data for new relations, which weakens feature learning and leads to overfitting. To mitigate this, we propose Pairwise Augmentation, a lightweight data enrichment strategy that generates diverse training

pairs without external resources. Combined with a dual-objective loss, our method effectively exploits augmented data to enhance representation learning and model generalization.

4.1.1. Mutual pairing augmentation

Our approach systematically expands the training set by generating composite, dual-label instances from the original data. This strategy creates challenging samples that require the model to disentangle superimposed contexts, thereby acting as a strong regularizer against overfitting in data-scarce scenarios.

Let the training set for a given task t be denoted as $D_t = \{(x_i, y_i)\}_{i=1}^{N_t}$, where $N_t = |D_t|$ is the number of available samples. We first define a set of all unique unordered pairs of indices from this set:

$$\mathcal{P}_t = \{\{i, j\} \mid 1 \le i < j \le N_t,\ y_i \neq y_j\} \tag{15}$$

We restrict $\mathcal{P}_t$ to the cross-relation subset: pairs with $y_i = y_j$ are inadmissible, since the dual-objective losses below ($\mathcal{L}_{\text{ML}}$ and $\mathcal{L}_{\text{CL}}$) are defined only for composites carrying two distinct labels. The total number of such pairs is upper-bounded by $L = \binom{N_t}{2} = \frac{N_t(N_t-1)}{2}$. For each index pair $\{i, j\} \in \mathcal{P}_t$, we generate a new augmented sample (x_k, y_k) via a concatenation operation, denoted by $\oplus$. The augmented input x_k is formed by concatenating the token sequences of the original texts, separated by a special `[SEP]` token:

$$x_k = x_i \oplus x_j \tag{16}$$

The corresponding augmented label y_k is constructed as an ordered pair of the original labels:

$$y_k = (y_i, y_j) \tag{17}$$

This process yields the final augmented dataset, D^t_{aug}, which is formally defined as:

$$D^t_{\text{aug}} = \{(x_i \oplus x_j, (y_i, y_j)) \mid \{i, j\} \in \mathcal{P}_t\} \tag{18}$$

This strategy transforms a dataset of size N_t into one of size $\mathcal{O}(N_t^2)$, significantly increasing data diversity. More importantly, it acts as a compositional, context-mixing regularization on the relational representation, reinforced by the explicit dual-objective loss introduced below. By requiring the model's self-attention mechanism to process two independent relations within a single forward pass, it must learn to disentangle contexts and build more robust, discriminative feature representations. This dynamic shares conceptual similarities with perturbation-based methods like Adversarial Training [56–58], ultimately improving generalization and mitigating overfitting. We invoke this analogy only in the broad sense of training with deliberately challenging composite samples, since PairAug does not perturb inputs along an adversarial gradient direction.

4.1.2. Dual-objective loss function

To fully exploit the augmented data, we introduce a dual-objective function that operates on two distinct label spaces: the original *single-label space* and the newly created *dual-label space*.

Margin loss for disentanglement (Single-label space). The first objective ensures that the model can correctly disentangle the two constituent relations within an augmented sample. Given an augmented sample $(x_k, [y_i, y_j])$, we extract the corresponding relation representations, z_k^i and z_k^j. The margin loss then pushes these two representations apart in the embedding space if their original labels are different. This forces the model to learn distinct features for each relation, even when they appear in a shared context. The loss is formulated as:

$$\mathcal{L}_{\text{ML}} = \frac{1}{N_{\text{diff}}} \sum_{k=1}^{L} \mathbb{I}_{y_i \neq y_j} \max(0, m - \text{sim}(z_k^i, z_k^j)) \tag{19}$$

where m is a predefined margin, $\text{sim}(\cdot, \cdot)$ is a similarity function (e.g., cosine similarity), $\mathbb{I}$ is the indicator function, and N_{diff} is the number of pairs with different labels. This loss enhances the discriminative power of the learned representations.

Contrastive loss for consolidation (Dual-label space). The second objective operates on the newly created dual-labels, treating each pair $[y_i, y_j]$ as a unique class. This encourages the model to consolidate its understanding of how different relations co-occur. Intuitively, two augmented samples that were created from the same set of original labels (e.g., $\{y_i, y_j\}$ and $\{y_j, y_i\}$) should have similar representations, while those created from different sets of labels should be pushed apart.

We use a supervised contrastive loss (InfoNCE) for this purpose. For each augmented sample k, we define its set of positives $P(k)$ as all other samples created from the same set of labels, $\{y_i, y_j\}$. The loss for sample k with mean representation $\bar{z}_k = \frac{1}{2}(z_k^i + z_k^j)$ is:

$$\mathcal{L}^k_{\text{Contra}} = -\frac{1}{|P(k)|} \sum_{p \in P(k)} \log \frac{\exp(\text{sim}(\bar{z}_k, \bar{z}_p)/\tau)}{\sum_{a \in N(k)} \exp(\text{sim}(\bar{z}_k, \bar{z}_a)/\tau)} \tag{20}$$

where $N(k)$ is the set of all other augmented samples in the batch (i.e negative examples), and τ is a temperature hyperparameter. The final contrastive loss is the average over all samples:

$$\mathcal{L}_{\text{CL}} = \frac{1}{L} \sum_{k=1}^{L} \mathcal{L}^k_{\text{Contra}} \tag{21}$$

This loss creates a structured embedding space for the dual-labels, further improving the model's generalization.

4.2. Enhancing robustness via guided sharpness-aware minimization

In this section, we present Guided Sharpness-Aware Minimization (GuidedSAM), an adaptive optimization framework driven by Salient Distillation. We first introduce Salient Distillation (Section 4.2.1), which identifies key informative tokens and measures representation deviation between consecutive tasks, producing an adaptive weight that reflects knowledge shift. Buiding on this, GuidedSAM leverages this signal to dynamically adjust the perturbation radius in SAM, enabling fine-grained control over the stability–plasticity trade-off.

4.2.1. Salient distillation

We propose **Salient Distillation (SD)**, a robust two-step strategy for calculating a *divergence metric*. Our core hypothesis is that effective knowledge retention hinges on first *identifying* the most influential tokens in a sequence and then *preserving* the intricate relational structure between them.

Identifying salient tokens via attention patterns. Not all tokens contribute equally to defining a semantic relation. To isolate the most critical tokens, we leverage the attention mechanism of the teacher model, $f_{\theta_{t-1}}$. In continual learning, the teacher model refers to the frozen state of the model after completing the previous task $t-1$. It serves as the stable benchmark to measure the relational divergence of the current model (f_θ). Following [59], we use the final transformer layer, as it typically captures the most global perspective on token importance.

Given an input sequence of length S, we first compute the average attention matrix $\mathbf{A} \in \mathbb{R}^{S\times S}$ by averaging the attention scores across all heads in the final layer. The importance score for the jth token is then calculated as the total attention it receives from all other tokens in the sequence:

$$\text{score}_j = \sum_{i=1}^{S} (\mathbf{A})_{i,j} \tag{22}$$

We then select the top-k tokens with the highest scores, forming a salient token subset denoted as $\mathcal{I}^{\text{top-}k}$. The choice of k is an important hyperparameter, which we analyze in Section 5.3.4.

Relational metric for salient embeddings. After identifying the most critical tokens, we must preserve their structural relationships. For this, we employ Relational Knowledge Distillation (RKD) [60], which excels at transferring the geometry of the embedding space. Crucially, we apply RKD metric exclusively to the embeddings of the salient token subset $\mathcal{I}^{\text{top-}k}$. This ensures that the divergence calculation process is focused, efficient, and less prone to noise from irrelevant tokens. RKD consists of two complementary components that capture different aspects of the relational structure:

- *Distance-wise Component (Δ_D)* This metric preserves the absolute distances between pairs of salient token embeddings. It encourages the student model (f_θ) to learn the same "distance structure" as the teacher ($f_{\theta_{t-1}}$), ensuring that tokens that were close or far apart in the teacher's representation space remain so in the student's. The distance potential, ψ_D, is defined as the normalized Euclidean distance:

$$\psi_D(h_i, h_j) = \frac{1}{\mu}\|h_i - h_j\|_2 \tag{23}$$

  where the full sequence of final-layer embeddings be $\mathbf{H} = [\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_S]$, where $\mathbf{h}_j \in \mathbb{R}^d$ is the embedding for the jth token (from either teacher or student) and μ is the average pairwise distance within the mini-batch, which provides normalization and training stability. The final distance component penalizes the difference between the teacher and student potentials over all pairs in the salient subset:

$$\Delta_D = \sum_{(i,j)\in\mathcal{I}^{\text{top-}k}} l_\delta(\psi_D(t_i, t_j), \psi_D(s_i, s_j)) \tag{24}$$

  where t_i and s_i are the teacher and student embeddings for a salient token, and l_δ is the Huber penalty function.

- *Angle-wise Component* (Δ_A) This metric captures higher-order, multi-object relationships by preserving the angles between vectors formed by triplets of salient embeddings. It transfers more complex structural properties, such as the orientation of one point relative to two others. The angular potential, ψ_A, is the cosine similarity between two difference vectors originating from a common point:

$$\psi_A(h_{ij}, h_{ik}) = \left\langle \frac{h_i - h_j}{\|h_i - h_j\|_2}, \frac{h_i - h_k}{\|h_i - h_k\|_2} \right\rangle \tag{25}$$

  The final angle component sums the differences in these angular potentials over all unique triplets in the salient subset:

$$\Delta_A = \sum_{(i,j,k)\in\mathcal{I}^{\text{top-}k}} l_\delta(\psi_A(t_{ij}, t_{ik}), \psi_A(s_{ij}, s_{ik})) \tag{26}$$

For both components, we use the Huber function, l_δ, which is more robust to outliers than a standard L_2 penalty:

$$l_\delta(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2 & \text{for } |y - \hat{y}| \leq \delta, \\ \delta(|y - \hat{y}| - \frac{1}{2}\delta) & \text{otherwise.} \end{cases} \tag{27}$$

where, δ is typically set to 1.

The final Salient Distillation score is a weighted sum of these two components, forming the *adaptive weight* used in our GuidedSAM module (Section 4.2):

$$\Delta_{\text{SD}} = \lambda_D \Delta_D + \lambda_A \Delta_A, \tag{28}$$

where λ_A and λ_D are hyperparameters that regulate the relative influence of each component. By focusing RKD on only the most salient tokens, our SD method creates a dense, low-noise, and structurally robust divergence metric, perfectly suited for the data-scarce FCRE environment.

4.2.2. Guided sharpness-aware minimization

To mitigate catastrophic forgetting while adapting to new few-shot relations, an effective optimizer should locate a solution that is both generalizable, corresponding to a flat region in the loss landscape, and consistent with prior knowledge. Although Sharpness-Aware Minimization (SAM) excels at discovering flat minima, its conventional design relies on a fixed-size l_p-norm ball to define the perturbation region. This static radius ρ is suboptimal for continual learning, as it cannot adjust to task-specific difficulty or the degree of divergence from previously learned knowledge. Inspired by trust-region optimization principles [31], we propose GuidedSAM that the perturbation radius ρ should be adaptive, functioning as a dynamic "trust region" that expands or contracts according to the degree of divergence between the model's current and previous relational representations. This adaptive mechanism constrains the optimizer from making excessively large updates that might reduce the current task loss but distort the underlying feature space and lead to significant forgetting of prior knowledge.

In the FCRE scenario, the model must carefully balance adaptation and retention. For semantically similar new relations, updates should remain conservative to preserve prior knowledge, whereas highly distinct relations require greater flexibility to explore the loss landscape. A fixed perturbation radius ρ cannot effectively accommodate this dynamic behavior.

We regulate the size of the trust region through the *adaptive weight* (Δ_{SD} in Eq. (28)), obtained from our Salient Distillation (Section 4.2.1). This distance, computed between the top-k salient token embeddings of the teacher and student models, provides a task-aware indicator of knowledge preservation.

- A **low** Δ_{SD} indicates that the student model's internal representations are still aligned with the teacher's. We can "trust" the model's current parameter region, so we use a **smaller** ρ to encourage fine-tuning and prevent unnecessary deviation.
- A **high** Δ_{SD} signals that the model is beginning to forget, with its representations diverging significantly. We "distrust" the current gradient direction and thus use a **larger** ρ to force the optimizer to search a wider neighborhood for a flatter, more stable minimum that can better accommodate both old and new knowledge.

This adaptive approach transforms SAM from a simple flatness-seeking algorithm into a knowledge-aware optimizer that actively counteracts forgetting. We formalize our GuidedSAM by dynamically setting the SAM neighborhood radius ρ at each optimization step t as a function of the relational distance Δ_{SD}:

$$\rho_t = \lambda_\rho \cdot \Delta_{\text{SD}}(\theta_t, \theta_{\text{teacher}}) \tag{29}$$

where λ_ρ is a hyperparameter that scales the influence of the relational distance. In practice, we clip ρ_t to a reasonable range $[\rho_{\min}, \rho_{\max}]$ to ensure stability.

Furthermore, to address the sensitivity of SAM to parameter scales, we adopt the normalization strategy from Adaptive Sharpness-Aware Minimization (ASAM) [29]. This ensures that perturbations are applied uniformly across the network, regardless of the magnitude of individual parameters. Combining our adaptive radius with ASAM and an l_2 weight decay regularizer, the final GuidedSAM optimization objective is:

$$\min_\theta \max_{\|\epsilon\|_2 \leq \rho_t} \mathcal{L}(\theta + \epsilon) + \frac{\eta}{2}\|\theta\|_2^2 \tag{30}$$

The two-step optimization process at each iteration t is as follows:

1. Ascent Step (*Find the worst-case perturbation*): For the standard l_2 ASAM case, this is:

$$\epsilon_t = \rho_t \cdot \frac{(\theta_t)^2 \odot \nabla\mathcal{L}(\theta_t)}{\|\theta_t \odot \nabla\mathcal{L}(\theta_t)\|_2} \tag{31}$$

2. Descent Step (*Update parameters*):

$$\theta_{t+1} = \theta_t - \alpha_t(\nabla\mathcal{L}(\theta_t + \epsilon_t) + \eta\theta_t) \tag{32}$$

Table 1
Performance comparison of existing FCRE SOTA methods and those when using SAM (i.e., +SAM).

	FewRel		TACRED	
FCRE	CPL	+SAM	CPL	+SAM
BERT	$75.53_{\pm 0.75}$	$75.71_{\pm 0.74}$	$66.60_{\pm 1.29}$	$66.68_{\pm 1.22}$
LLM2Vec	$82.51_{\pm 0.67}$	$81.99_{\pm 0.65}$	$75.32_{\pm 1.88}$	$76.31_{\pm 1.48}$

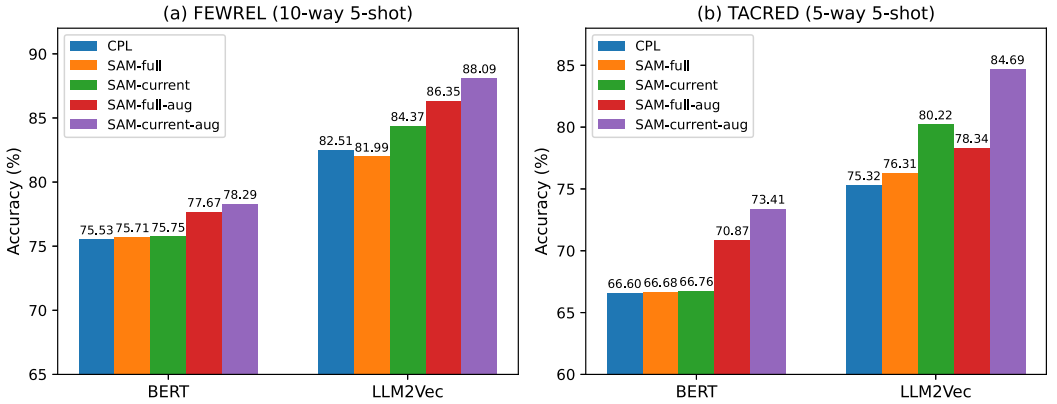


Fig. 3. Applications of SAM in FCRE models. *"SAM-full"* indicates the case we apply SAM on all data $\mathcal{D}^t$ from the current task and memory buffer, *"SAM-current"* is when we apply SAM only on the data of the current task.

where α_t is the learning rate and η is the weight decay coefficient. Following the findings in Section 4.3, we apply the GuidedSAM update only to the data from the current task, excluding it from the memory replay buffer to maximize stability. By making the optimizer explicitly aware of relational knowledge drift, GuidedSAM encourages the model to find minima that are not only flat but also lie within a region of the parameter space that respects previously learned feature structures, thereby boosting generalization and robustly alleviating catastrophic forgetting.

Finally, combining Eqs. (19) and (21), the overall training objective can be formulated as:

$$\mathcal{L} = \mathcal{L}_0 + \beta_1 \mathcal{L}_{ML} + \beta_2 \mathcal{L}_{CL} \tag{33}$$

where $\mathcal{L}_0$ denotes the base objective of the underlying framework (i.e., Eqs. (3) and (8) for SIRUS and CPL, respectively), while β_1 and β_2 are hyperparameters that balance the contributions of each loss component.

4.3. Understanding SAM under data scarcity in FCRE

Sharpness-Aware Minimization (SAM) [54] is well-established as a technique for improving model generalization. However, its efficacy in the challenging, data-scarce environment of Few-shot Continual Learning has not been systematically studied. In this section, we conduct a targeted investigation into the behavior of SAM in FCRE, providing key insights that motivate the design of our proposed framework.

The challenge of naive SAM application in FCRE. Our initial investigation involved a straightforward application of SAM to the entire training set at each continual learning step t, which includes both the new task data, $\mathcal{D}^t$, and the replayed memory buffer, $\mathcal{M}^{t-1}$. The optimization objective was:

$$\min_{\theta} \max_{\|\epsilon\| \leq \rho} \mathcal{L}(\mathcal{M}^{t-1} \cup \mathcal{D}^t; \theta + \epsilon) \tag{34}$$

The results, presented in Table 1, were counter-intuitive. For the FCRE task, applying SAM naively provided inconsistent gains and, in several cases, led to significant performance degradation. For instance, with the CPL_MI method on the TACRED dataset, this approach resulted in a 2% drop in final accuracy. This finding suggests that SAM is not a "plug-and-play" solution for FCRE and its benefits are not guaranteed.

The data scarcity hypothesis: isolating the memory buffer. We hypothesize that the observed performance degradation arises from applying SAM's perturbation-based optimization to the extremely sparse memory buffer. Since the buffer often contains only a single instance per old relation, it yields noisy and unreliable gradients for sharpness estimation, which may misguide the optimizer.

To validate this hypothesis, we design a hybrid optimization strategy: SAM is applied exclusively to the current task data $\mathcal{D}^t$ (which provides richer supervision), while a standard optimization objective is used for the sparse memory buffer $\mathcal{M}^{t-1}$. The resulting objective is defined as:

Table 2
Hyperparameter configuration for SIRUS + PRAGAS.

Hyperparameter	BERT	LLM
Current-task training epochs	8	8
Memory training epochs	6	6
Learning rate	1e-5	1e-5
Encoder output dimension	768	4096
BERT input maximum sequence length	256	–
Top-k for Salient Distillation	10	10
λ_A, λ_D	1 , 1	1, 1
λ_1	1	1
λ_2	2	2
λ_3	0.5	0.5
λ_ρ	3	3

$$\min_{\theta} \left(\mathcal{L}(\mathcal{M}^{t-1}; \theta) + \max_{\|\epsilon\| \leq \rho} \mathcal{L}(\mathcal{D}^t; \theta + \epsilon) \right) \tag{35}$$

As shown in Fig. 3, this hybrid approach consistently recovered the benefits of SAM, leading to improved performance across different models. This result strongly supports our hypothesis: **SAM's effectiveness in continual learning is highly sensitive to data scarcity, particularly within the memory replay mechanism.**

Pairwise augmentation as a principled solution. The preceding analysis reveals a clear requirement: for SAM to be effective on replayed data, the memory buffer must be sufficiently rich and diverse. This observation directly motivates our **PairAug** strategy.

Our augmentation method transforms the sparse memory buffer into a dense and varied training set. At task t, with n_m samples in memory and n_c samples for the new task, our method generates a quadratically larger set of training instances for both past and current relations. The number of samples representing old tasks is effectively increased by a factor proportional to $\frac{n_m - 1}{2} + n_c$, providing a much richer signal for the SAM optimizer.

As evidenced by our main results in Table 5, when SAM is combined with our PairAug, it yields significant and consistent performance gains. Unlike the hybrid approach, which is a workaround, our strategy provides a principled solution by directly addressing SAM's core weakness in the FCRE context: the lack of data. This synergy between data augmentation and advanced optimization is a cornerstone of our proposed PRAGAS framework.

5. Experiments

In this section, we detail the experimental setup, results, and analyses conducted to validate the effectiveness of our proposed approach on the FCRE task, comparing it against several state-of-the-art baselines. While our primary focus is on FCRE, we further extend the evaluation to Continual Event Detection (CED) to assess the generalizability of the method, with CED-related settings and results provided in the Appendix.

5.1. Experimental setup

Datasets. We evaluate our proposed framework on two widely used benchmarks for FCRE: **FewRel** [61] and **TACRED** [62]. The FewRel dataset contains 100 relations with 70,000 samples, from which we follow the setting of Qin and Joty [9] by selecting 80 relations divided into 8 tasks, each comprising 10 relations (10-way). The first task T^1 includes 100 samples per relation, while the remaining tasks adopt a 5-shot configuration. TACRED consists of 42 relations and 106,264 instances derived from Newswire and Web documents. Consistent with Qin and Joty [9], we exclude the "*no_relation*" class and partition the remaining 41 relations into 8 tasks, where the first task T^1 includes 6 relations with 100 samples each, and subsequent tasks contain 5 relations (5-way) under 5-shot settings.

Baselines. This section summarizes several state-of-the-art approaches in Few-Shot Continual Relation Extraction (FCRE) that serve as benchmark baselines in our evaluations. **SCKD** [18] employs structured knowledge distillation and contrastive learning with pseudo-samples to preserve knowledge from earlier tasks and enhance relation representation. **RP-CRE** [63] mitigates forgetting by generating relation prototypes through K-means clustering on stored samples, allowing the model to adjust embeddings of new samples efficiently. **CRL** [51] maintains consistent relation embeddings using supervised contrastive learning and knowledge distillation during memory replay, effectively reducing catastrophic forgetting. **CRECL** [64] enriches instance representations using support examples and cross-domain knowledge, demonstrating improved performance in low-data scenarios. **ERDA** [9] introduces embedding space regularization and self-supervised data augmentation to learn new relations with minimal data while preventing forgetting.

Additional baselines include **ConPL** [10], which combines prototype-based classification, memory enhancement, and consistent learning, augmented with prompt learning and focal loss to reduce confusion between similar classes. **CPL** [23] applies contrastive prompt learning with margin-based objectives and memory augmentation using ChatGPT to reduce forgetting and overfitting, while

Table 3
Hyperparameter configuration for CPL + PRAGAS.

Hyperparameter	BERT	LLM
Current-task training epochs	8	8
Memory training epochs	6	8
Learning rate	1e-5	1e-5
Encoder output dimension	768	4096
BERT input maximum sequence length	256	–
Margin (m) for Margin Loss	1.0	1.0
Top-k for Salient Distillation	10	10
λ_A, λ_D	1 , 1	1, 1
β_1	0.25	0.25
β_2	0.25	0.25
λ_ρ	6	6

Table 4
Contributions of PairAug and GuidedSAM. "w.o X" indicates the ablation setting where component X is excluded from the PRAGAS framework.

Method	T^1	T^2	T^3	T^4	T^5	T^6	T^7	T^8	Avg
FewRel (10-way 5-shot)									
CPL + PRAGAS	$95.10_{\pm 0.17}$	$87.63_{\pm 3.10}$	$82.06_{\pm 1.57}$	$77.90_{\pm 2.35}$	$76.08_{\pm 2.41}$	$72.83_{\pm 2.01}$	$70.28_{\pm 1.42}$	$68.16_{\pm 0.56}$	$78.76_{\pm 0.68}$
w.o all	$94.99_{\pm 0.18}$	$83.22_{\pm 2.87}$	$77.35_{\pm 1.96}$	$74.10_{\pm 2.37}$	$71.96_{\pm 2.97}$	$69.71_{\pm 2.55}$	$67.25_{\pm 1.38}$	$65.63_{\pm 1.02}$	$75.53_{\pm 0.75}$
w.o GuidedSAM	$94.89_{\pm 0.31}$	$86.50_{\pm 3.18}$	$81.86_{\pm 1.46}$	$78.26_{\pm 2.22}$	$76.15_{\pm 2.50}$	$72.73_{\pm 2.34}$	$70.16_{\pm 1.69}$	$68.04_{\pm 0.52}$	$78.57_{\pm 0.71}$
w.o PairAug	$95.11_{\pm 0.43}$	$84.78_{\pm 3.19}$	$77.65_{\pm 1.74}$	$74.14_{\pm 2.55}$	$72.07_{\pm 3.47}$	$69.36_{\pm 2.13}$	$67.42_{\pm 1.40}$	$65.49_{\pm 0.42}$	$75.75_{\pm 0.78}$
TACRED (5-way 5-shot)									
CPL + PRAGAS	$86.87_{\pm 0.42}$	$84.32_{\pm 6.34}$	$79.32_{\pm 3.04}$	$75.98_{\pm 4.99}$	$72.12_{\pm 5.82}$	$70.32_{\pm 5.02}$	$66.92_{\pm 4.43}$	$64.00_{\pm 3.86}$	$74.98_{\pm 1.62}$
w.o all	$86.58_{\pm 0.40}$	$80.75_{\pm 5.58}$	$71.31_{\pm 4.33}$	$66.53_{\pm 2.36}$	$62.22_{\pm 4.17}$	$58.78_{\pm 4.66}$	$54.23_{\pm 2.95}$	$52.42_{\pm 1.84}$	$66.60_{\pm 1.29}$
w.o GuidedSAM	$86.04_{\pm 0.80}$	$81.96_{\pm 5.45}$	$75.64_{\pm 2.16}$	$72.36_{\pm 3.89}$	$66.97_{\pm 4.40}$	$65.57_{\pm 3.40}$	$62.37_{\pm 2.80}$	$60.33_{\pm 2.39}$	$71.41_{\pm 1.22}$
w.o PairAug	$86.87_{\pm 0.42}$	$80.64_{\pm 5.46}$	$71.68_{\pm 3.26}$	$67.16_{\pm 3.14}$	$60.94_{\pm 5.06}$	$59.43_{\pm 3.70}$	$55.18_{\pm 2.27}$	$52.19_{\pm 2.02}$	$66.76_{\pm 1.24}$

CPL_MI [22] maximizes mutual information between language model heads and classifiers to preserve prior knowledge and improve representations. Finally, **SIRUS** [25] addresses the challenge of similar classes by dynamically clustering relations based on their descriptions. For consistency, we reproduce ConPL results using the same settings as SCKD and CPL, as the original evaluation strategy is not directly applicable to continual learning scenarios.

Backbone encoders. To obtain input representations as described in Section 3.2, we evaluate our method across a range of encoder backbones. In the primary experiments, we adopt the standard `bert-base-uncased` model [50]. Further showcase the adaptability, we explore more advanced encoders with Large Language Models (LLMs), including the BGE sentence-embedding model [33] and three variants of LLM2Vec [34] based on **Llama2** [65], **Llama3** [66], and **Mistral**.

Evaluation metric. To assess model performance, we employ the final average accuracy. Specifically, the average accuracy after completing task $\mathcal{T}^j$ is computed as:

$$ACC[j] = \frac{1}{j}\sum_{i=1}^{j} ACC[j,i]$$

where $ACC[j,i]$ denotes the accuracy on the test set of task $\mathcal{T}^i$ evaluated after the model has been trained on task $\mathcal{T}^j$. For all reported results, we perform 6 independent runs using different random seeds and present the mean along with the corresponding standard deviation.

Configurations. All experiments were conducted on an Ubuntu Server 22.04.3 LTS environment. Specifically, the BERT-based experiments were run on an NVIDIA Tesla P100 GPU with 16GB of memory, while experiments using LLM2Vec and BGE backbones were executed on an NVIDIA A100 GPU equipped with 80GB of VRAM. Details regarding the hyperparameter configurations are provided in Tables 2 and 3.

5.2. Comprehensive evaluation of PRAGAS

In this section, we conduct a comprehensive evaluation of PRAGAS by comparing it with several state-of-the-art approaches in FCRE. Our experiments demonstrate the versatility of PRAGAS in substantially enhancing existing frameworks and models across different architectures and scales. Beyond performance gains, we further analyze the computational efficiency and overhead introduced by PRAGAS, highlighting its practicality for real-world continual learning scenarios.

Table 5
Accuracy (%) of different BERT-based methods after training for each task on TACRED and FewRel in 5-shot settings. We highlight the rows corresponding to our method. The best result in each group is in **bold**. §Results of ConPL are reproduced with the same settings as other models.

Method	T^1	T^2	T^3	T^4	T^5	T^6	T^7	T^8	Avg
FewRel (10-way 5-shot)									
RP-CRE	$93.97_{\pm 0.64}$	$76.05_{\pm 2.36}$	$71.36_{\pm 2.83}$	$69.32_{\pm 3.98}$	$64.95_{\pm 3.09}$	$61.99_{\pm 2.09}$	$60.59_{\pm 1.87}$	$59.57_{\pm 1.13}$	$69.73_{\pm 0.87}$
CRL	$94.68_{\pm 0.33}$	$80.73_{\pm 2.91}$	$73.82_{\pm 2.77}$	$70.26_{\pm 3.18}$	$66.62_{\pm 2.74}$	$63.28_{\pm 2.49}$	$60.96_{\pm 2.63}$	$59.27_{\pm 1.32}$	$71.20_{\pm 0.87}$
CRECL	$93.93_{\pm 0.22}$	$82.55_{\pm 6.95}$	$74.13_{\pm 3.59}$	$69.33_{\pm 3.87}$	$66.51_{\pm 4.05}$	$64.60_{\pm 1.92}$	$62.97_{\pm 1.46}$	$59.99_{\pm 0.65}$	$71.75_{\pm 1.24}$
ERDA	$92.43_{\pm 0.32}$	$64.52_{\pm 2.11}$	$50.31_{\pm 3.32}$	$44.92_{\pm 3.77}$	$39.75_{\pm 3.34}$	$36.36_{\pm 3.12}$	$34.34_{\pm 1.83}$	$31.96_{\pm 1.91}$	$49.32_{\pm 0.95}$
SCKD	$94.77_{\pm 0.35}$	$82.83_{\pm 2.61}$	$76.21_{\pm 1.61}$	$72.19_{\pm 1.33}$	$70.61_{\pm 2.24}$	$67.15_{\pm 1.96}$	$64.86_{\pm 1.35}$	$62.98_{\pm 0.88}$	$73.95_{\pm 0.60}$
ConPL§	$\mathbf{95.18_{\pm 0.73}}$	$79.63_{\pm 1.27}$	$74.54_{\pm 1.13}$	$71.27_{\pm 0.85}$	$68.35_{\pm 0.86}$	$63.86_{\pm 2.03}$	$64.74_{\pm 1.39}$	$62.46_{\pm 1.54}$	$72.50_{\pm 0.46}$
CPL	$94.99_{\pm 0.18}$	$83.22_{\pm 2.87}$	$77.35_{\pm 1.96}$	$74.10_{\pm 2.37}$	$71.96_{\pm 2.97}$	$69.71_{\pm 2.55}$	$67.25_{\pm 1.38}$	$65.63_{\pm 1.02}$	$75.53_{\pm 0.75}$
CPL_MI	$94.69_{\pm 0.70}$	$85.58_{\pm 1.88}$	$80.12_{\pm 2.45}$	$75.71_{\pm 2.28}$	$73.90_{\pm 1.80}$	$70.72_{\pm 0.91}$	$68.42_{\pm 1.77}$	$66.27_{\pm 1.58}$	$76.93_{\pm 0.62}$
Sirus	$94.64_{\pm 0.19}$	$85.79_{\pm 2.23}$	$80.08_{\pm 1.41}$	$76.68_{\pm 2.44}$	$74.70_{\pm 2.88}$	$72.04_{\pm 1.64}$	$69.71_{\pm 1.40}$	$67.76_{\pm 0.63}$	$77.68_{\pm 0.64}$
CPL + Pragas	$95.10_{\pm 0.17}$	$87.63_{\pm 3.10}$	$\mathbf{82.06_{\pm 1.57}}$	$77.90_{\pm 2.35}$	$76.08_{\pm 2.41}$	$72.83_{\pm 2.01}$	$70.28_{\pm 1.42}$	$68.16_{\pm 0.56}$	$78.76_{\pm 0.68}$
Sirus + Pragas	$95.10_{\pm 0.26}$	$\mathbf{87.78_{\pm 1.93}}$	$81.80_{\pm 1.45}$	$\mathbf{80.06_{\pm 1.70}}$	$\mathbf{78.19_{\pm 2.43}}$	$\mathbf{74.49_{\pm 2.78}}$	$\mathbf{71.92_{\pm 0.67}}$	$\mathbf{70.08_{\pm 0.73}}$	$\mathbf{79.93_{\pm 0.60}}$
TACRED (5-way 5-shot)									
RP-CRE	$87.32_{\pm 1.76}$	$74.90_{\pm 6.13}$	$67.88_{\pm 4.31}$	$60.02_{\pm 5.37}$	$53.26_{\pm 4.67}$	$50.72_{\pm 7.62}$	$46.21_{\pm 5.29}$	$44.48_{\pm 3.74}$	$60.60_{\pm 1.81}$
CRL	$88.32_{\pm 1.26}$	$76.30_{\pm 7.48}$	$69.76_{\pm 5.89}$	$61.93_{\pm 2.55}$	$54.68_{\pm 3.12}$	$50.92_{\pm 4.45}$	$47.00_{\pm 3.78}$	$44.27_{\pm 2.51}$	$61.65_{\pm 1.53}$
CRECL	$87.09_{\pm 2.50}$	$78.09_{\pm 5.74}$	$61.93_{\pm 4.89}$	$55.60_{\pm 5.78}$	$53.42_{\pm 2.99}$	$51.91_{\pm 2.95}$	$47.55_{\pm 3.38}$	$45.53_{\pm 1.96}$	$60.14_{\pm 1.42}$
ERDA	$81.88_{\pm 1.97}$	$53.68_{\pm 6.31}$	$40.36_{\pm 3.35}$	$36.17_{\pm 3.65}$	$30.14_{\pm 3.96}$	$22.61_{\pm 3.13}$	$22.29_{\pm 1.32}$	$19.42_{\pm 2.31}$	$38.32_{\pm 1.25}$
SCKD	$88.42_{\pm 0.83}$	$79.35_{\pm 4.13}$	$70.61_{\pm 3.16}$	$66.78_{\pm 4.29}$	$60.47_{\pm 3.05}$	$58.05_{\pm 3.84}$	$54.41_{\pm 3.47}$	$52.11_{\pm 3.15}$	$66.28_{\pm 1.20}$
ConPL§	$\mathbf{88.77_{\pm 0.84}}$	$69.64_{\pm 1.93}$	$57.50_{\pm 2.48}$	$52.15_{\pm 1.59}$	$58.19_{\pm 2.31}$	$55.01_{\pm 3.12}$	$52.88_{\pm 3.66}$	$50.97_{\pm 3.41}$	$60.64_{\pm 0.91}$
CPL	$86.58_{\pm 0.40}$	$80.75_{\pm 5.58}$	$71.31_{\pm 4.33}$	$66.53_{\pm 2.36}$	$62.22_{\pm 4.17}$	$58.78_{\pm 4.66}$	$54.23_{\pm 2.95}$	$52.42_{\pm 1.84}$	$66.60_{\pm 1.29}$
CPL_MI	$85.67_{\pm 0.80}$	$82.54_{\pm 2.98}$	$75.12_{\pm 3.67}$	$70.65_{\pm 2.75}$	$66.79_{\pm 2.18}$	$65.17_{\pm 2.48}$	$61.25_{\pm 1.52}$	$59.48_{\pm 3.53}$	$70.83_{\pm 0.94}$
Sirus	$87.22_{\pm 0.48}$	$83.66_{\pm 5.46}$	$75.83_{\pm 2.56}$	$72.24_{\pm 4.31}$	$67.95_{\pm 3.42}$	$65.92_{\pm 3.29}$	$61.97_{\pm 4.10}$	$59.95_{\pm 1.94}$	$71.84_{\pm 1.24}$
CPL + Pragas	$86.87_{\pm 0.42}$	$84.32_{\pm 6.34}$	$79.32_{\pm 3.04}$	$75.98_{\pm 4.99}$	$72.12_{\pm 5.82}$	$70.32_{\pm 5.02}$	$66.92_{\pm 4.43}$	$64.00_{\pm 3.86}$	$74.98_{\pm 1.62}$
Sirus + Pragas	$87.47_{\pm 0.26}$	$\mathbf{86.04_{\pm 6.16}}$	$\mathbf{79.76_{\pm 5.63}}$	$\mathbf{77.26_{\pm 4.13}}$	$\mathbf{74.70_{\pm 4.56}}$	$\mathbf{73.80_{\pm 4.22}}$	$\mathbf{71.54_{\pm 2.17}}$	$\mathbf{69.83_{\pm 1.66}}$	$\mathbf{77.55_{\pm 1.44}}$

Effectiveness of PairAug and GuidedSAM. The results in Table 4 clearly highlight the effectiveness of both PairAug and GuidedSAM in our framework. The most striking observation is the pivotal role of PairAug: on the TACRED dataset, removing this component (w.o PairAug) leads to a drastic drop in average accuracy from 74.98% to 66.76%, nearly eliminating all performance gains over the baseline. While PairAug establishes a strong foundation by expanding the data space, the integration of GuidedSAM further enhances performance, as evidenced by a reduction from 74.98% to 71.41% when it is removed. This indicates that although additional data contributes substantially, the optimization strategy guided by our method plays an equally crucial role in exploiting it effectively. Importantly, this dependence on PairAug also explains the FewRel observation in Table 4: GuidedSAM alone, without PairAug, yields only marginal improvement over the CPL baseline (75.75% vs. 75.53%), and the same pattern holds on TACRED (66.76% vs. 66.60%). As established in Section 4.3, SAM-based optimization requires a sufficiently dense data signal to locate effective flat minima, a condition the sparse single-exemplar memory buffer alone cannot satisfy. PairAug supplies this density, which is why the meaningful gains appear in combination (+3.23 on FewRel and +8.38 on TACRED over the CPL baseline) rather than from GuidedSAM in isolation. As shown in Fig. 3 and Table 1, naively applying SAM yields inconsistent performance across datasets and backbones, confirming that SAM alone does not generalize well under data-scarce FCRE settings. Consistent and substantial improvements across datasets are observed only when SAM is integrated with PairAug through our GuidedSAM strategy, further demonstrating the complementary and synergistic nature of the two proposed components and validating the rationale of our overall design.

Pushing the state of the art in FCRE with Pragas. As shown in Table 5, integrating Pragas into existing FCRE frameworks leads to substantial improvements across all tasks. On the FewRel (10-way 5-shot) benchmark, CPL + Pragas achieves an average accuracy of 78.76%, outperforming the original CPL (75.53%) by more than 3.2 percentage points, while Sirus + Pragas attains the best overall performance of 79.93%, exceeding Sirus by 2.25 points. Similarly, on the more challenging TACRED (5-way 5-shot) setting, the integration of Pragas elevates the performance of both CPL and Sirus, reaching 74.98% and 77.55% average accuracy, surpassing their respective baselines by over 4% and 3.7%. These consistent gains across both datasets confirm the effectiveness of Pragas in mitigating data scarcity and improving continual adaptation. The enhanced frameworks exhibit stronger generalization to emerging relations while maintaining stability in previously learned ones. Notably, the improvements remain consistent across all training stages, demonstrating the robustness of our method in handling FCRE scenarios.

Statistical significance of the gains. To verify that these improvements are not within-noise fluctuations, we performed paired significance tests on the six-seed runs underlying Table 5, using the per-seed AVG values matched across methods. We report both a parametric paired t-test and a non-parametric Wilcoxon signed-rank test (two-sided), along with Cohen's d_z effect size. The four cross-method comparisons against the CPL and Sirus baselines are summarized in Table 6. The comparison with the smallest margin, FewRel CPL + Pragas vs Sirus, is significant under the paired t-test ($p = 0.0069$); of its six matched-seed differences, five are positive

Table 6
Paired significance tests on the four cross-method comparisons in Table 5, over six matched seeds. Significance markers: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Comparison	Δ	Paired t-test	Wilcoxon signed-rank	Cohen d_z
FewRel: CPL + PRAGAS vs CPL	+3.23	$t = 11.66$, $p = 8.2\times10^{-5}$ ***	$W = 0$, $p = 0.031$ *	4.76
FewRel: CPL + PRAGAS vs SIRUS	+1.08	$t = 4.42$, $p = 0.0069$ **	$W = 1$, $p = 0.063$	1.80
TACRED: CPL + PRAGAS vs CPL	+8.38	$t = 10.73$, $p = 1.2\times10^{-4}$ ***	$W = 0$, $p = 0.031$ *	4.38
TACRED: CPL + PRAGAS vs SIRUS	+3.14	$t = 3.42$, $p = 0.0188$ *	$W = 0$, $p = 0.031$ *	1.40

Table 7
Paired tests on SIRUS + PRAGAS vs SIRUS (Table 5), over six matched seeds. Markers: * $p < 0.05$, *** $p < 0.001$.

Comparison	Δ	Paired t-test	Wilcoxon signed-rank	Cohen d_z
FewRel: SIRUS + PRAGAS vs SIRUS	+2.25	$t = 10.61$, $p = 1.3\times10^{-4}$ ***	$W = 0$, $p = 0.031$ *	4.33
TACRED: SIRUS + PRAGAS vs SIRUS	+5.71	$t = 3.99$, $p = 0.010$ *	$W = 0$, $p = 0.031$ *	1.63

Table 8
Final average accuracy and average forgetting F [67] across six matched seeds. Bold marks the best value per column.

Method	FewRel Avg (↑)	FewRel F (↓)	TACRED Avg (↑)	TACRED F (↓)
CPL	$75.53_{\pm0.75}$	$\mathbf{11.87_{\pm1.23}}$	$66.60_{\pm1.29}$	$24.18_{\pm2.57}$
CPL + PRAGAS	$\mathbf{78.76_{\pm0.68}}$	$15.86_{\pm1.14}$	$\mathbf{74.98_{\pm1.62}}$	$\mathbf{16.92_{\pm4.07}}$

and one is a small negative (−0.06%), yielding a Wilcoxon $p = 0.063$ just above the conventional cutoff. We note that the smallest two-sided p-value attainable by Wilcoxon at $n = 6$ is $1/2^5 = 0.031$, so the test has limited resolution at this sample size. The remaining three comparisons are significant under both tests at $p < 0.05$. Under per-family Bonferroni correction ($\alpha/4 = 0.0125$), three of the four comparisons remain significant on the t-test; the marginal exception is TACRED CPL + PRAGAS vs SIRUS ($p = 0.0188$), which is significant at uncorrected $\alpha = 0.05$ but falls just above the Bonferroni threshold.

We further test the two SIRUS + PRAGAS vs SIRUS comparisons (Table 7). Both are statistically significant under both tests, with a very large matched-seed effect size on FewRel ($d_z = 4.33$, t-test $p < 0.001$) and a large effect size on TACRED ($d_z = 1.63$, t-test $p = 0.010$). Combined with the cross-framework gains over CPL above, this confirms that PRAGAS provides a real, framework-agnostic improvement rather than a noise-level fluctuation.

Alleviating the forgetting phenomenon. To provide a deeper analysis of how our proposed PRAGAS mitigates catastrophic forgetting, we visualize the task-wise accuracy evolution throughout the continual learning process, as shown in Fig. 4. Each subplot corresponds to a task, and the curves trace the model's performance on that task as new tasks are incrementally introduced. Across both datasets, **FewRel** and **TACRED**, a clear pattern emerges: while the baseline CPL model (blue) exhibits a consistent degradation in performance as new tasks arrive, the enhanced CPL+PRAGAS (orange) maintains notably higher accuracy at nearly every stage. This stability demonstrates that PRAGAS effectively preserves prior knowledge while continuing to learn new relations. The performance gap between the two models becomes increasingly evident in later tasks (e.g., T^6 - T^8), where conventional continual learning models typically experience severe forgetting. In contrast, CPL+PRAGAS shows a slower and smoother decline in accuracy, confirming that our combination of PairAug and GuidedSAM contributes to improved knowledge retention and reduced interference from new task updates. These observations reinforce that PRAGAS not only improves final average accuracy but also delivers consistent learning dynamics, sustaining performance across tasks and effectively mitigating catastrophic forgetting throughout the continual training process.

Beyond the task-wise accuracy curves in Fig. 4, we report the standard Average Forgetting F of Chaudhry et al. [67] to quantify stability against a widely used metric. It is computed from the per-task accuracy matrix as

$$F = \frac{1}{T-1}\sum_{i=1}^{T-1}\left(\max_{j\in[i,T-1]} ACC[j,i] - ACC[T,i]\right), \tag{36}$$

where $ACC[j,i]$ denotes the accuracy on task i after training on task j and $T = 8$. We report F alongside the final Average Accuracy (Avg) as the two complementary axes of stability and plasticity recommended by Chaudhry et al. [67]; results across the six matched seeds are summarized in Table 8.

Stability–plasticity trade-off. On TACRED, CPL + PRAGAS improves both axes simultaneously: F drops by 30.1% relative (24.18% → 16.92%) while Avg gains +8.38 points over CPL (Table 6). On FewRel, where CPL already retains relatively well (F = 11.87%), PRAGAS exhibits a measured trade-off: F rises by 3.99 points while Avg rises by a statistically significant 3.23 points. This dataset-dependent pattern is a direct consequence of GuidedSAM's adaptive perturbation radius, which scales with the Salient Distillation deviation Δ_{SD} between consecutive tasks (Section 4.2). On the noisier TACRED relation set Δ_{SD} is large, so the radius approaches ρ_{max} and GuidedSAM aggressively flattens the loss landscape; on the cleaner FewRel relations Δ_{SD} stays small, the radius is pinned

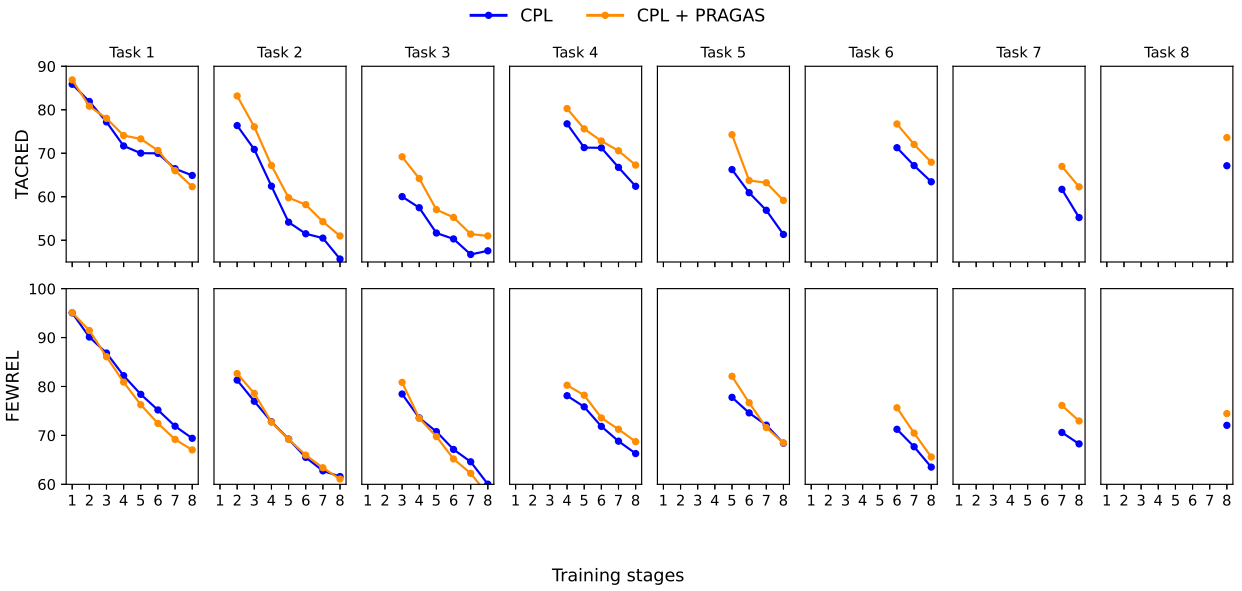


Fig. 4. Accuracy (%) on each task overtime.

near $\rho_{\min}$, and the stabilization contribution is weaker, precisely the regime in which Section 4.3 predicts flat-minima search has limited room to operate. Reducing F further on FewRel would necessarily constrain the additional learning that PairAug provides, approaching the degenerate frozen-baseline regime where $F = 0$ but Avg stagnates [67]. The reported trade-off therefore favors plasticity exactly where the resulting accuracy gain is significant, consistent with the design intent of an adaptive sharpness-aware optimizer rather than a uniformly conservative one.

Broad effectiveness and adaptability of PRAGAS. As shown in Table 5, PRAGAS demonstrates strong orthogonality and universal applicability across diverse FCRE frameworks, consistently yielding substantial performance improvements. To rigorously evaluate its robustness and generalizability, we conducted an extensive ablation study in which the standard BERT encoder was replaced with a variety of more powerful backbone models (Table 9). These include state-of-the-art sentence embedding encoders such as **BGE**, as well as large language model–based representations (e.g., Mistral and Llama) obtained through the **LLM2Vec** framework. Our first key observation is that employing these advanced encoders significantly elevates baseline performance. For example, when using the SIRUS framework, replacing the BERT backbone with BGE raises the average accuracy on FewRel from 77.68% to 86.26%, while adopting a Mistral-based backbone improves TACRED accuracy from 71.84% to 78.11%. These improvements substantially raise the performance ceiling, making further progress more challenging. However, despite their superior representational power, these models still suffer from catastrophic forgetting, as evidenced by consistent performance degradation in later tasks, which is aligned with prior findings on causal LLMs in FCRE [22].

The next objective is to determine whether the benefits of PRAGAS persist when applied to these stronger backbones. The results compellingly indicate that PRAGAS not only remains compatible but also exhibits strong improvement with advanced encoders. On the **TACRED** dataset, integrating PRAGAS into the BGE-based SIRUS framework increases average accuracy from 76.79% to 84.86%, representing a remarkable improvement of $+\mathbf{8.07\%}$. Similarly, on **FewRel**, incorporating PRAGAS into the Llama3 backbone yields a gain of $+\mathbf{3.08\%}$ (from 85.88% to 88.96%). These findings confirm that PRAGAS is a universal, model-agnostic framework: rather than offering marginal improvements, it fundamentally strengthens continual learning across diverse architectures by addressing the twin challenges of data scarcity and knowledge retention through principled data augmentation and guided optimization.

Table 9
Accuracy (%) of methods using LLM-based backbones after training for each task. The baseline results of CPL are obtained from [22].

Backbone	Method	T^1	T^2	T^3	T^4	T^5	T^6	T^7	T^8	Avg
FewRel (10-way 5-shot)										
BGE	CPL	$96.52_{\pm 0.26}$	$89.88_{\pm 3.33}$	$84.3_{\pm 1.97}$	$81.5_{\pm 2.94}$	$79.05_{\pm 4.02}$	$77.27_{\pm 3.49}$	$75.6_{\pm 2.99}$	$73.25_{\pm 2.48}$	$82.17_{\pm 1.02}$
	SIRUS	$96.90_{\pm 0.34}$	$91.14_{\pm 1.83}$	$87.94_{\pm 1.46}$	$86.39_{\pm 2.11}$	$84.62_{\pm 2.22}$	$82.82_{\pm 1.96}$	$80.9_{\pm 0.69}$	$79.38_{\pm 0.48}$	$86.26_{\pm 0.55}$
	CPL + PRAGAS	$97.05_{\pm 0.28}$	$91.26_{\pm 2.64}$	$86.14_{\pm 2.04}$	$84.20_{\pm 2.43}$	$82.35_{\pm 3.12}$	$80.48_{\pm 2.86}$	$78.91_{\pm 2.01}$	$76.72_{\pm 1.80}$	$84.14_{\pm 0.88}$
	SIRUS + PRAGAS	$\mathbf{97.20_{\pm 0.31}}$	$\mathbf{92.34_{\pm 1.65}}$	$\mathbf{89.26_{\pm 1.53}}$	$\mathbf{87.92_{\pm 2.00}}$	$\mathbf{86.40_{\pm 2.01}}$	$\mathbf{84.89_{\pm 1.88}}$	$\mathbf{83.05_{\pm 0.82}}$	$\mathbf{81.80_{\pm 0.65}}$	$\mathbf{87.86_{\pm 0.62}}$
Mistral	CPL	$96.6_{\pm 0.22}$	$88.75_{\pm 2.63}$	$84.39_{\pm 2.65}$	$82.46_{\pm 2.08}$	$80.38_{\pm 1.93}$	$78.06_{\pm 1.18}$	$75.41_{\pm 1.90}$	$74.00_{\pm 1.30}$	$82.51_{\pm 0.67}$
	SIRUS	$96.13_{\pm 0.31}$	$89.74_{\pm 2.69}$	$86.10_{\pm 2.41}$	$84.25_{\pm 2.25}$	$81.96_{\pm 2.81}$	$79.79_{\pm 2.56}$	$77.75_{\pm 2.09}$	$76.96_{\pm 1.15}$	$84.09_{\pm 0.77}$
	CPL + PRAGAS	$96.92_{\pm 0.25}$	$91.10_{\pm 2.40}$	$87.12_{\pm 2.15}$	$85.06_{\pm 2.02}$	$83.74_{\pm 1.88}$	$81.62_{\pm 1.50}$	$79.58_{\pm 1.42}$	$78.22_{\pm 1.21}$	$85.17_{\pm 0.70}$
	SIRUS + PRAGAS	$\mathbf{97.08_{\pm 0.28}}$	$\mathbf{92.38_{\pm 2.20}}$	$\mathbf{88.64_{\pm 1.91}}$	$\mathbf{86.90_{\pm 1.95}}$	$\mathbf{85.28_{\pm 1.92}}$	$\mathbf{83.46_{\pm 1.43}}$	$\mathbf{81.85_{\pm 1.16}}$	$\mathbf{80.46_{\pm 0.99}}$	$\mathbf{86.76_{\pm 0.65}}$
Llama2	CPL	$96.78_{\pm 0.37}$	$88.91_{\pm 3.38}$	$84.90_{\pm 2.93}$	$81.99_{\pm 2.21}$	$79.20_{\pm 3.28}$	$77.60_{\pm 2.40}$	$75.57_{\pm 3.20}$	$74.12_{\pm 1.23}$	$82.38_{\pm 0.91}$
	SIRUS	$95.58_{\pm 0.24}$	$87.93_{\pm 2.54}$	$83.22_{\pm 1.51}$	$81.76_{\pm 1.01}$	$81.23_{\pm 2.25}$	$79.07_{\pm 1.69}$	$76.86_{\pm 1.63}$	$75.98_{\pm 0.58}$	$82.70_{\pm 0.57}$
	CPL + PRAGAS	$97.00_{\pm 0.29}$	$90.34_{\pm 2.64}$	$86.26_{\pm 2.28}$	$83.85_{\pm 1.92}$	$81.92_{\pm 2.41}$	$80.03_{\pm 1.84}$	$78.12_{\pm 1.66}$	$76.84_{\pm 1.07}$	$84.04_{\pm 0.75}$
	SIRUS + PRAGAS	$\mathbf{97.14_{\pm 0.30}}$	$\mathbf{91.82_{\pm 2.31}}$	$\mathbf{87.92_{\pm 2.04}}$	$\mathbf{85.64_{\pm 1.80}}$	$\mathbf{84.03_{\pm 2.10}}$	$\mathbf{82.18_{\pm 1.68}}$	$\mathbf{80.60_{\pm 1.32}}$	$\mathbf{79.35_{\pm 0.92}}$	$\mathbf{86.08_{\pm 0.63}}$
Llama3	CPL	$97.37_{\pm 0.15}$	$87.96_{\pm 2.66}$	$83.02_{\pm 1.34}$	$79.78_{\pm 2.78}$	$78.09_{\pm 3.09}$	$75.95_{\pm 1.87}$	$74.65_{\pm 1.60}$	$73.19_{\pm 1.11}$	$81.25_{\pm 0.72}$
	SIRUS	$96.80_{\pm 0.18}$	$91.04_{\pm 2.43}$	$87.36_{\pm 1.49}$	$85.25_{\pm 1.48}$	$84.28_{\pm 2.69}$	$82.46_{\pm 1.67}$	$81.03_{\pm 1.42}$	$78.82_{\pm 0.98}$	$85.88_{\pm 0.60}$
	CPL + PRAGAS	$\mathbf{97.45_{\pm 0.21}}$	$90.12_{\pm 2.55}$	$86.40_{\pm 1.72}$	$84.08_{\pm 2.31}$	$82.74_{\pm 2.41}$	$80.83_{\pm 1.74}$	$79.35_{\pm 1.55}$	$78.02_{\pm 1.07}$	$84.87_{\pm 0.68}$
	SIRUS + PRAGAS	$97.02_{\pm 0.25}$	$\mathbf{92.18_{\pm 2.01}}$	$\mathbf{89.49_{\pm 1.72}}$	$\mathbf{88.30_{\pm 1.68}}$	$\mathbf{87.71_{\pm 1.96}}$	$\mathbf{86.75_{\pm 1.53}}$	$\mathbf{85.74_{\pm 1.32}}$	$\mathbf{84.52_{\pm 0.92}}$	$\mathbf{88.96_{\pm 0.64}}$
TACRED (5-way 5-shot)										
BGE	CPL	$88.43_{\pm 0.86}$	$85.51_{\pm 5.27}$	$77.76_{\pm 5.19}$	$75.79_{\pm 4.57}$	$74.23_{\pm 2.93}$	$71.97_{\pm 4.47}$	$70.68_{\pm 5.11}$	$67.52_{\pm 5.37}$	$76.49_{\pm 1.58}$
	SIRUS	$88.33_{\pm 0.14}$	$85.72_{\pm 3.46}$	$78.80_{\pm 2.80}$	$75.23_{\pm 3.62}$	$73.62_{\pm 2.18}$	$72.34_{\pm 1.64}$	$70.18_{\pm 3.57}$	$70.07_{\pm 2.74}$	$76.79_{\pm 0.97}$
	CPL + PRAGAS	$88.91_{\pm 0.44}$	$87.66_{\pm 4.93}$	$82.61_{\pm 4.26}$	$80.32_{\pm 3.97}$	$78.54_{\pm 3.25}$	$76.82_{\pm 3.28}$	$74.93_{\pm 3.05}$	$73.18_{\pm 2.91}$	$80.37_{\pm 1.42}$
	SIRUS + PRAGAS	$\mathbf{90.06_{\pm 0.48}}$	$\mathbf{88.96_{\pm 4.45}}$	$\mathbf{85.78_{\pm 3.87}}$	$\mathbf{83.85_{\pm 3.09}}$	$\mathbf{83.91_{\pm 2.52}}$	$\mathbf{83.51_{\pm 0.76}}$	$\mathbf{82.14_{\pm 1.32}}$	$\mathbf{80.66_{\pm 1.57}}$	$\mathbf{84.86_{\pm 0.93}}$
Mistral	CPL	$88.56_{\pm 0.58}$	$83.57_{\pm 5.25}$	$75.54_{\pm 6.82}$	$74.82_{\pm 5.28}$	$72.55_{\pm 4.99}$	$71.13_{\pm 6.44}$	$69.05_{\pm 5.94}$	$67.36_{\pm 4.67}$	$75.32_{\pm 1.88}$
	SIRUS	$88.24_{\pm 0.23}$	$83.29_{\pm 5.02}$	$79.12_{\pm 3.98}$	$76.92_{\pm 3.74}$	$75.26_{\pm 3.24}$	$75.31_{\pm 1.40}$	$73.64_{\pm 4.97}$	$73.06_{\pm 3.23}$	$78.11_{\pm 1.27}$
	CPL + PRAGAS	$85.97_{\pm 5.63}$	$87.80_{\pm 4.56}$	$83.56_{\pm 4.96}$	$81.52_{\pm 3.40}$	$79.52_{\pm 5.79}$	$78.52_{\pm 4.53}$	$75.98_{\pm 4.49}$	$75.75_{\pm 4.22}$	$81.08_{\pm 1.68}$
	SIRUS + PRAGAS	$\mathbf{89.52_{\pm 0.29}}$	$\mathbf{88.75_{\pm 3.91}}$	$\mathbf{86.08_{\pm 2.66}}$	$\mathbf{86.52_{\pm 2.51}}$	$\mathbf{84.88_{\pm 3.63}}$	$\mathbf{84.62_{\pm 2.28}}$	$\mathbf{83.22_{\pm 1.55}}$	$\mathbf{81.72_{\pm 1.22}}$	$\mathbf{85.66_{\pm 0.89}}$
Llama2	CPL	$87.37_{\pm 1.85}$	$82.74_{\pm 9.54}$	$77.49_{\pm 7.52}$	$77.29_{\pm 4.49}$	$72.75_{\pm 6.28}$	$73.37_{\pm 4.57}$	$70.08_{\pm 6.01}$	$68.35_{\pm 5.02}$	$76.18_{\pm 2.14}$
	SIRUS	$89.62_{\pm 0.31}$	$87.07_{\pm 7.02}$	$78.98_{\pm 4.58}$	$76.04_{\pm 3.28}$	$74.64_{\pm 3.15}$	$74.14_{\pm 2.39}$	$70.96_{\pm 1.77}$	$70.88_{\pm 0.59}$	$77.79_{\pm 1.25}$
	CPL + PRAGAS	$\mathbf{90.14_{\pm 0.36}}$	$88.12_{\pm 6.35}$	$83.46_{\pm 5.12}$	$81.54_{\pm 3.98}$	$80.22_{\pm 4.12}$	$79.01_{\pm 3.43}$	$77.14_{\pm 2.81}$	$75.89_{\pm 2.30}$	$81.69_{\pm 1.39}$
	SIRUS + PRAGAS	$89.94_{\pm 0.41}$	$\mathbf{88.17_{\pm 6.26}}$	$\mathbf{84.32_{\pm 4.35}}$	$\mathbf{84.22_{\pm 3.01}}$	$\mathbf{82.32_{\pm 3.58}}$	$\mathbf{81.63_{\pm 3.04}}$	$\mathbf{79.31_{\pm 1.88}}$	$\mathbf{79.39_{\pm 1.69}}$	$\mathbf{83.66_{\pm 1.22}}$
Llama3	CPL	$88.75_{\pm 0.59}$	$81.18_{\pm 9.26}$	$76.14_{\pm 4.36}$	$76.16_{\pm 5.60}$	$72.14_{\pm 5.85}$	$71.35_{\pm 5.41}$	$69.99_{\pm 5.21}$	$69.70_{\pm 5.36}$	$75.68_{\pm 2.00}$
	SIRUS	$87.76_{\pm 0.61}$	$85.85_{\pm 3.97}$	$82.19_{\pm 4.19}$	$77.61_{\pm 2.67}$	$74.86_{\pm 3.41}$	$75.67_{\pm 3.03}$	$74.42_{\pm 4.02}$	$73.97_{\pm 3.71}$	$79.04_{\pm 1.20}$
	CPL + PRAGAS	$89.13_{\pm 0.45}$	$86.24_{\pm 6.82}$	$83.56_{\pm 4.75}$	$81.88_{\pm 4.26}$	$80.13_{\pm 3.90}$	$79.24_{\pm 3.22}$	$77.92_{\pm 3.01}$	$76.61_{\pm 2.94}$	$81.59_{\pm 1.31}$
	SIRUS + PRAGAS	$\mathbf{89.52_{\pm 0.15}}$	$\mathbf{88.66_{\pm 4.92}}$	$\mathbf{85.51_{\pm 3.51}}$	$\mathbf{84.31_{\pm 3.52}}$	$\mathbf{83.56_{\pm 3.42}}$	$\mathbf{83.62_{\pm 1.90}}$	$\mathbf{81.74_{\pm 1.73}}$	$\mathbf{80.59_{\pm 1.22}}$	$\mathbf{84.69_{\pm 1.04}}$

5.3. Ablation study

5.3.1. Comparison with simpler augmentation baselines

A natural question is whether PairAug's gains stem from its specific pairing structure or simply from expanding the training set. The $\mathcal{O}(N_t^2)$ expansion conflates two effects: having more training data, and the dual-label structure of paired inputs. To disentangle them, we compare PairAug against four simpler augmentation baselines on TACRED (CPL + PRAGAS framework), all matched to PairAug on the number of augmented samples per generation pass so that the data-quantity axis is held fixed across rows. The four baselines bracket the two design axes: *random duplication* (no perturbation) tests whether sheer volume suffices; *random duplication with WordNet synonym replacement* adds a word-level lexical perturbation; *cross-sample sentence concatenation* preserves PairAug's input-level concatenation but discards the dual-label structure (only the first sample's relation is kept, so training uses a single-label objective); and *EDA word-level random insertion* [68] injects WordNet-derived synonyms at random positions. None of these produces the dual-label composites that drive PairAug.

The results in Table 10 are decisive: at a matched augmented-sample budget, none of the four simpler baselines improves over the CPL baseline (66.60), and three of them slightly degrade it, whereas PairAug alone raises accuracy to 71.41 and the full PRAGAS reaches 74.98. Since the CPL baseline already incorporates ChatGPT-generated synthetic samples, this is not a "no data" versus "more data" comparison; the +8.38-point gain from adding PairAug therefore cannot be attributed to data volume alone, isolating the contribution of the pairing structure itself.

5.3.2. Impact of loss components

Our proposed PRAGAS employs a dual-objective optimization scheme composed of two complementary loss functions: the Margin Loss ($\mathcal{L}_{\text{ML}}$, Eq. (19)) designed for relation disentanglement, and the Contrastive Loss ($\mathcal{L}_{\text{CL}}$, Eq. (21)) aimed at representation consolidation. To examine the individual contribution and necessity of each component, we conducted an ablation study in which each loss term was selectively removed. The quantitative results are summarized in Table 11.

Table 10
Matched-budget ablation comparing PairAug against simpler augmentation baselines on TACRED. The four simpler-augmentation rows are matched to PairAug on the number of augmented samples per generation pass, isolating the contribution of the pairing structure from raw data volume. All rows use the CPL backbone, six seeds per cell.

Variant	TACRED Avg
CPL baseline (w.o all)	66.60
CPL + random duplication (no perturbation)	65.01
CPL + random duplication + WordNet synonym replacement	64.84
CPL + cross-sample sentence concatenation (single label)	65.71
CPL + EDA word-level random insertion	65.70
CPL + PairAug only (w.o GuidedSAM)	71.41
CPL + GuidedSAM only (w.o PairAug)	66.76
CPL + PRAGAS (full)	**74.98**

Table 11
Ablation study of loss components. Performance comparison when removing the margin loss ($\mathcal{L}_{\text{ML}}$) or the contrastive loss ($\mathcal{L}_{\text{CL}}$) from the PRAGAS framework.

Method	T^1	T^2	T^3	T^4	T^5	T^6	T^7	T^8	Avg
FewRel (10-way 5-shot)									
CPL + PRAGAS	$95.10_{\pm 0.17}$	$87.63_{\pm 3.10}$	$82.06_{\pm 1.57}$	$77.90_{\pm 2.35}$	$76.08_{\pm 2.41}$	$72.83_{\pm 2.01}$	$70.28_{\pm 1.42}$	$68.16_{\pm 0.56}$	$78.76_{\pm 0.68}$
w.o $\mathcal{L}_{\text{ML}}$	$95.10_{\pm 0.44}$	$87.45_{\pm 3.35}$	$81.75_{\pm 1.45}$	$77.42_{\pm 2.33}$	$75.31_{\pm 2.55}$	$72.15_{\pm 1.85}$	$69.82_{\pm 1.83}$	$67.26_{\pm 1.09}$	$78.28_{\pm 0.72}$
w.o $\mathcal{L}_{\text{CL}}$	$95.10_{\pm 0.44}$	$87.36_{\pm 3.39}$	$81.70_{\pm 1.60}$	$77.71_{\pm 2.25}$	$75.65_{\pm 2.32}$	$72.05_{\pm 1.64}$	$69.69_{\pm 1.62}$	$67.39_{\pm 0.97}$	$78.33_{\pm 0.70}$
TACRED (5-way 5-shot)									
CPL + PRAGAS	$86.87_{\pm 0.42}$	$84.32_{\pm 6.34}$	$79.32_{\pm 3.04}$	$75.98_{\pm 4.99}$	$72.12_{\pm 5.82}$	$70.32_{\pm 5.02}$	$66.92_{\pm 4.43}$	$64.00_{\pm 3.86}$	$74.98_{\pm 1.62}$
w.o $\mathcal{L}_{\text{ML}}$	$86.87_{\pm 0.42}$	$84.15_{\pm 6.38}$	$79.38_{\pm 3.21}$	$75.74_{\pm 5.04}$	$71.65_{\pm 5.86}$	$69.72_{\pm 5.39}$	$66.40_{\pm 4.21}$	$64.13_{\pm 4.25}$	$74.76_{\pm 1.66}$
w.o $\mathcal{L}_{\text{CL}}$	$86.87_{\pm 0.42}$	$83.61_{\pm 6.88}$	$78.01_{\pm 2.30}$	$73.93_{\pm 4.27}$	$70.40_{\pm 6.19}$	$69.21_{\pm 5.16}$	$65.66_{\pm 4.78}$	$63.74_{\pm 4.61}$	$73.93_{\pm 1.68}$

The results clearly indicate that both objectives are indispensable to the overall effectiveness of PRAGAS. Removing either one leads to measurable performance degradation, underscoring their complementary roles.

- Impact of Margin Loss: Eliminating the Margin Loss (w.o $\mathcal{L}_{\text{ML}}$) weakens the model's ability to disentangle relational signals within augmented pairs, resulting in reduced average accuracy on both FewRel (from 78.76% to 78.28%) and TACRED (from 74.98% to 74.76%). This demonstrates that enforcing a margin between paired relations is crucial for learning distinct, non-overlapping representations, particularly important in the presence of overlapping or semantically similar relation types.

- Impact of Contrastive Loss: Removing the Contrastive Loss (w.o $\mathcal{L}_{\text{CL}}$) produces an even stronger decline, most notably on the more complex TACRED dataset, where average accuracy drops by over 1% (from 74.98% to 73.93%). This term plays a pivotal role in shaping a structured embedding space that anchors semantically similar dual-label pairs while maintaining inter-class separability. Without this consolidation mechanism, the model struggles to fully utilize the relational diversity introduced through augmentation, leading to poorer generalization and higher confusion between related event types.

Overall, this analysis validates the dual-objective design of PRAGAS. The $\mathcal{L}_{\text{ML}}$ and $\mathcal{L}_{\text{CL}}$ address two complementary dimensions of continual relational learning: disentanglement and consolidation. Together, they enable the model to not only learn robust and discriminative representations from augmented samples but also to maintain coherent relational structures across evolving tasks, thereby ensuring both stability and adaptability in continual learning settings.

Because these per-component differences are small in absolute terms, we test their significance using the same paired protocol as above (six matched seeds, per-seed AVG). This is a *confirmatory* ablation: each loss term is expected to contribute, so a detectable difference is the desired outcome. As shown in Table 12, three of the four comparisons are significant under both tests, and although the magnitudes are modest the matched-seed effect sizes are large to very large ($d_z = 1.57$ to 2.16), showing the per-seed gains are systematic rather than seed-to-seed noise. The exception is TACRED without $\mathcal{L}_{\text{ML}}$, whose small magnitude ($\Delta = 0.22\%$) falls below detectability at $n = 6$: on the noisier TACRED relation set $\mathcal{L}_{\text{CL}}$ carries the dominant statistical share while $\mathcal{L}_{\text{ML}}$ provides a complementary effect that is detectable on FewRel but not separable from noise on TACRED. These within-method tests are reported as descriptive evidence and are not multiplicity-corrected.

5.3.3. *Exploring the impact of SAM variants*

A central contribution of this work is the introduction of GuidedSAM (Section 4.2), a specialized SAM-based technique designed specifically for continual learning scenarios. In this section, we present a comprehensive ablation study comparing GuidedSAM with several representative SAM variants integrated within the PRAGAS framework. The compared methods include the original SAM, as well as its notable variants: ASAM [29], ESAM [30], GCSAM [69], FriendlySAM [70], and LookbehindASAM [55]. Unlike these

Table 12
Paired significance tests on the loss-component ablation (Table 11), six matched seeds. Markers: * $p < 0.05$, ** $p < 0.01$, ns = not significant.

Comparison	Δ	Paired t-test	Wilcoxon	Cohen d_z
FewRel Full vs w.o $\mathcal{L}_{\mathrm{ML}}$	+0.47	$t = 5.28$, $p = 0.003$ **	$W = 0$, $p = 0.031$ *	2.16
FewRel Full vs w.o $\mathcal{L}_{\mathrm{CL}}$	+0.42	$t = 3.85$, $p = 0.012$ *	$W = 0$, $p = 0.031$ *	1.57
TACRED Full vs w.o $\mathcal{L}_{\mathrm{ML}}$	+0.22	$t = 1.39$, $p = 0.225$ ns	$W = 4$, $p = 0.219$ ns	0.57
TACRED Full vs w.o $\mathcal{L}_{\mathrm{CL}}$	+1.05	$t = 4.35$, $p = 0.007$ **	$W = 0$, $p = 0.031$ *	1.78

Table 13
Comparative performance of GuidedSAM and existing SAM variants on two benchmarks. Results are reported across two base continual learning frameworks (CPL and SIRUS).

Base Method	SAM Variant	T^1	T^2	T^3	T^4	T^5	T^6	T^7	T^8	Avg
FewRel (10-way 5-shot)										
CPL	SAM	$95.17_{\pm 0.35}$	$86.85_{\pm 3.07}$	$81.95_{\pm 1.74}$	$77.72_{\pm 1.66}$	$75.60_{\pm 2.56}$	$72.19_{\pm 2.30}$	$69.69_{\pm 1.56}$	$67.17_{\pm 1.03}$	$78.29_{\pm 0.69}$
	ASAM	$95.10_{\pm 0.44}$	$87.31_{\pm 3.27}$	$81.93_{\pm 1.41}$	$77.45_{\pm 2.53}$	$75.70_{\pm 2.62}$	$72.41_{\pm 2.03}$	$69.62_{\pm 2.01}$	$67.57_{\pm 0.83}$	$78.39_{\pm 0.74}$
	ESAM	$95.09_{\pm 0.21}$	$86.87_{\pm 3.01}$	$81.88_{\pm 2.06}$	$77.86_{\pm 2.52}$	$75.90_{\pm 2.68}$	$72.43_{\pm 1.98}$	$69.69_{\pm 1.29}$	$67.47_{\pm 0.62}$	$78.40_{\pm 0.72}$
	GCSAM	$94.97_{\pm 0.34}$	$86.87_{\pm 3.23}$	$82.04_{\pm 1.76}$	$77.88_{\pm 2.29}$	$75.50_{\pm 2.77}$	$72.04_{\pm 2.40}$	$69.70_{\pm 1.54}$	$67.55_{\pm 0.89}$	$78.32_{\pm 0.74}$
	FriendlySAM	$95.07_{\pm 0.45}$	$86.68_{\pm 2.90}$	$82.08_{\pm 1.58}$	$77.58_{\pm 2.21}$	$75.34_{\pm 2.23}$	$72.22_{\pm 2.01}$	$69.39_{\pm 1.79}$	$67.04_{\pm 1.32}$	$78.18_{\pm 0.68}$
	LookbehindASAM	$90.77_{\pm 0.98}$	$83.81_{\pm 2.73}$	$78.36_{\pm 1.66}$	$75.23_{\pm 2.63}$	$73.09_{\pm 2.82}$	$69.48_{\pm 2.15}$	$67.28_{\pm 1.92}$	$65.36_{\pm 1.48}$	$75.42_{\pm 0.76}$
	GuidedSAM	$95.07_{\pm 0.17}$	$\mathbf{87.63}_{\pm \mathbf{3.10}}$	$82.06_{\pm 1.57}$	$\mathbf{77.90}_{\pm \mathbf{2.35}}$	$\mathbf{76.08}_{\pm \mathbf{2.41}}$	$\mathbf{72.83}_{\pm \mathbf{2.01}}$	$\mathbf{70.28}_{\pm \mathbf{1.42}}$	$\mathbf{68.16}_{\pm \mathbf{0.56}}$	$\mathbf{78.76}_{\pm \mathbf{0.68}}$
SIRUS	SAM	$94.64_{\pm 0.28}$	$86.40_{\pm 1.91}$	$80.14_{\pm 1.29}$	$78.63_{\pm 2.01}$	$77.06_{\pm 2.05}$	$73.33_{\pm 2.59}$	$69.76_{\pm 0.68}$	$67.88_{\pm 0.25}$	$78.48_{\pm 0.57}$
	ASAM	$93.40_{\pm 3.19}$	$82.85_{\pm 7.04}$	$80.33_{\pm 3.43}$	$79.82_{\pm 1.72}$	$77.99_{\pm 2.36}$	$\mathbf{75.50}_{\pm \mathbf{2.40}}$	$72.12_{\pm 1.75}$	$\mathbf{71.83}_{\pm \mathbf{1.20}}$	$79.23_{\pm 1.19}$
	ESAM	$\mathbf{94.74}_{\pm \mathbf{0.16}}$	$85.35_{\pm 2.66}$	$80.93_{\pm 1.49}$	$78.99_{\pm 1.88}$	$77.38_{\pm 2.21}$	$73.73_{\pm 2.16}$	$70.52_{\pm 0.82}$	$67.88_{\pm 0.37}$	$78.69_{\pm 0.60}$
	GCSAM	$94.71_{\pm 0.09}$	$85.40_{\pm 2.60}$	$81.16_{\pm 2.03}$	$79.34_{\pm 2.01}$	$77.34_{\pm 2.27}$	$73.52_{\pm 2.43}$	$70.79_{\pm 0.91}$	$68.67_{\pm 0.30}$	$78.87_{\pm 0.65}$
	FriendlySAM	$94.91_{\pm 0.19}$	$85.30_{\pm 2.69}$	$80.98_{\pm 2.52}$	$79.13_{\pm 2.08}$	$77.56_{\pm 2.25}$	$74.16_{\pm 2.53}$	$71.28_{\pm 0.39}$	$68.97_{\pm 0.45}$	$79.04_{\pm 0.68}$
	LookbehindASAM	$93.18_{\pm 3.06}$	$82.06_{\pm 6.68}$	$80.20_{\pm 3.40}$	$79.23_{\pm 2.27}$	$78.02_{\pm 2.02}$	$75.20_{\pm 1.93}$	$71.86_{\pm 1.58}$	$71.65_{\pm 1.08}$	$78.93_{\pm 1.13}$
	GuidedSAM	$95.07_{\pm 0.26}$	$\mathbf{87.78}_{\pm \mathbf{1.93}}$	$\mathbf{81.80}_{\pm \mathbf{1.45}}$	$\mathbf{80.06}_{\pm \mathbf{1.70}}$	$\mathbf{78.19}_{\pm \mathbf{2.43}}$	$74.49_{\pm 2.78}$	$\mathbf{71.92}_{\pm \mathbf{0.67}}$	$70.08_{\pm 0.73}$	$\mathbf{79.92}_{\pm \mathbf{0.60}}$
TACRED (5-way 5-shot)										
CPL	SAM	$86.58_{\pm 0.35}$	$82.76_{\pm 7.19}$	$78.36_{\pm 3.25}$	$74.11_{\pm 5.23}$	$69.18_{\pm 5.26}$	$68.23_{\pm 4.34}$	$64.73_{\pm 4.50}$	$63.35_{\pm 4.40}$	$73.41_{\pm 1.66}$
	ASAM	$86.64_{\pm 0.24}$	$83.16_{\pm 5.10}$	$78.34_{\pm 2.83}$	$74.72_{\pm 4.71}$	$70.81_{\pm 6.56}$	$69.78_{\pm 5.53}$	$66.67_{\pm 4.23}$	$64.75_{\pm 4.07}$	$74.36_{\pm 1.60}$
	ESAM	$86.71_{\pm 0.39}$	$82.62_{\pm 7.27}$	$76.57_{\pm 4.83}$	$72.94_{\pm 5.13}$	$68.62_{\pm 4.22}$	$67.84_{\pm 3.68}$	$64.49_{\pm 4.18}$	$62.47_{\pm 2.98}$	$72.78_{\pm 1.58}$
	GCSAM	$86.65_{\pm 0.61}$	$82.80_{\pm 6.94}$	$77.78_{\pm 3.66}$	$73.74_{\pm 5.30}$	$68.97_{\pm 4.72}$	$68.54_{\pm 3.89}$	$65.81_{\pm 3.17}$	$62.88_{\pm 3.22}$	$73.40_{\pm 1.52}$
	FriendlySAM	$86.61_{\pm 0.53}$	$83.29_{\pm 6.17}$	$77.86_{\pm 3.98}$	$73.50_{\pm 5.86}$	$70.24_{\pm 5.30}$	$68.69_{\pm 4.25}$	$65.63_{\pm 4.05}$	$64.08_{\pm 2.47}$	$73.74_{\pm 1.57}$
	LookbehindASAM	$\mathbf{87.25}_{\pm \mathbf{0.46}}$	$82.66_{\pm 8.88}$	$77.46_{\pm 3.95}$	$73.84_{\pm 5.50}$	$69.89_{\pm 5.82}$	$68.92_{\pm 4.98}$	$66.41_{\pm 3.64}$	$63.83_{\pm 3.13}$	$73.78_{\pm 1.80}$
	GuidedSAM	$86.87_{\pm 0.42}$	$\mathbf{84.32}_{\pm \mathbf{6.34}}$	$\mathbf{79.32}_{\pm \mathbf{3.04}}$	$\mathbf{75.98}_{\pm \mathbf{4.99}}$	$\mathbf{72.12}_{\pm \mathbf{5.82}}$	$\mathbf{70.32}_{\pm \mathbf{5.02}}$	$\mathbf{66.92}_{\pm \mathbf{4.43}}$	$\mathbf{64.00}_{\pm \mathbf{3.86}}$	$\mathbf{74.98}_{\pm \mathbf{1.62}}$
SIRUS	SAM	$87.06_{\pm 0.75}$	$83.47_{\pm 6.77}$	$77.59_{\pm 5.18}$	$74.00_{\pm 3.88}$	$70.52_{\pm 6.43}$	$69.74_{\pm 5.41}$	$67.33_{\pm 3.15}$	$66.77_{\pm 2.75}$	$74.56_{\pm 1.66}$
	ASAM	$87.47_{\pm 0.26}$	$85.90_{\pm 6.75}$	$79.96_{\pm 5.49}$	$77.66_{\pm 3.78}$	$74.03_{\pm 4.53}$	$73.49_{\pm 4.55}$	$70.76_{\pm 2.78}$	$68.92_{\pm 2.83}$	$77.27_{\pm 1.52}$
	ESAM	$87.18_{\pm 0.73}$	$84.03_{\pm 6.55}$	$78.02_{\pm 4.77}$	$74.13_{\pm 4.28}$	$70.40_{\pm 5.47}$	$69.66_{\pm 5.25}$	$66.76_{\pm 3.17}$	$66.56_{\pm 3.63}$	$74.59_{\pm 1.61}$
	GCSAM	$87.25_{\pm 0.73}$	$84.08_{\pm 5.77}$	$77.88_{\pm 5.54}$	$75.20_{\pm 3.55}$	$71.09_{\pm 6.49}$	$70.37_{\pm 5.23}$	$68.06_{\pm 2.84}$	$67.21_{\pm 2.44}$	$75.14_{\pm 1.58}$
	FriendlySAM	$87.03_{\pm 0.82}$	$84.15_{\pm 6.76}$	$77.57_{\pm 4.84}$	$74.04_{\pm 3.38}$	$70.33_{\pm 6.93}$	$69.99_{\pm 5.62}$	$67.94_{\pm 3.45}$	$66.65_{\pm 3.33}$	$74.71_{\pm 1.69}$
	LookbehindASAM	$\mathbf{87.50}_{\pm \mathbf{0.19}}$	$\mathbf{86.21}_{\pm \mathbf{5.35}}$	$\mathbf{80.34}_{\pm \mathbf{3.38}}$	$76.75_{\pm 3.60}$	$73.88_{\pm 4.66}$	$73.32_{\pm 3.66}$	$70.64_{\pm 2.38}$	$69.44_{\pm 2.50}$	$77.26_{\pm 1.25}$
	GuidedSAM	$87.47_{\pm 0.26}$	$86.04_{\pm 6.16}$	$79.76_{\pm 5.63}$	$\mathbf{77.26}_{\pm \mathbf{4.13}}$	$\mathbf{74.70}_{\pm \mathbf{4.56}}$	$\mathbf{73.80}_{\pm \mathbf{4.22}}$	$\mathbf{71.54}_{\pm \mathbf{2.17}}$	$\mathbf{69.83}_{\pm \mathbf{1.66}}$	$\mathbf{77.55}_{\pm \mathbf{1.44}}$

existing formulations, GuidedSAM uniquely combines the adaptive gradient scaling of ASAM with our proposed Salient Distillation mechanism, enabling task-aware optimization that balances model plasticity and stability during continual learning.

Table 13 presents a detailed comparison of GuidedSAM with six prominent variants of Sharpness-Aware Minimization (SAM) across two benchmark datasets (FewRel and TACRED) and two base continual learning frameworks (CPL and SIRUS). We first selected ASAM as the foundation for GuidedSAM, as it consistently achieves the strongest and most stable results among all existing SAM variants. Compared to the original SAM, ASAM demonstrates superior average accuracy across both datasets and frameworks; for example, in FewRel with the SIRUS baseline, ASAM improves the average accuracy from 78.48% to 79.23%. This improvement can be attributed to ASAM's adaptive scaling mechanism, which dynamically adjusts perturbation magnitudes based on gradient norms, leading to smoother and more context-aware optimization.

Building on ASAM's adaptive property, GuidedSAM further integrates our proposed Salient Distillation module to enhance task-aware regularization. The results reveal that GuidedSAM consistently surpasses all other SAM variants, achieving the best overall average accuracy and the lowest forgetting rates across settings. On FewRel, GuidedSAM achieves 78.76% and 79.92% average accuracy for CPL and SIRUS, respectively, outperforming the next best variant by +0.35% and +0.69%. Similarly, on TACRED, GuidedSAM reaches 74.98% (CPL) and 77.55% (SIRUS), marking clear improvements over both ASAM and other variants. These findings confirm that while existing SAM variants improve generalization through flat-minima optimization, they remain limited in addressing task interference in continual learning. In contrast, GuidedSAM leverages both gradient adaptivity and salient knowledge distillation, yielding more balanced plasticity–stability trade-offs and robust performance across heterogeneous tasks and datasets.

Table 14
Ablation study on top-k in Salient Distillation.

Top-k	T^1	T^2	T^3	T^4	T^5	T^6	T^7	T^8	Avg
FewRel (10-way 5-shot)									
3	$95.10_{\pm 0.44}$	$87.47_{\pm 3.36}$	$81.63_{\pm 1.50}$	$77.66_{\pm 2.20}$	$75.57_{\pm 2.58}$	$72.04_{\pm 1.94}$	$69.94_{\pm 1.96}$	$67.51_{\pm 1.07}$	$78.37_{\pm 0.73}$
5	$95.10_{\pm 0.44}$	$87.47_{\pm 3.36}$	$81.65_{\pm 1.15}$	$77.66_{\pm 2.30}$	$75.55_{\pm 2.74}$	$72.06_{\pm 1.85}$	$69.47_{\pm 1.82}$	$67.28_{\pm 1.04}$	$78.28_{\pm 0.72}$
10	$95.10_{\pm 0.17}$	$87.63_{\pm 3.10}$	$82.06_{\pm 1.57}$	$77.90_{\pm 2.35}$	$76.08_{\pm 2.41}$	$72.83_{\pm 2.01}$	$70.28_{\pm 1.42}$	$68.16_{\pm 0.56}$	$\mathbf{78.76_{\pm 0.68}}$
15	$95.10_{\pm 0.44}$	$87.47_{\pm 3.36}$	$81.61_{\pm 1.17}$	$77.61_{\pm 2.18}$	$75.60_{\pm 2.37}$	$71.95_{\pm 1.63}$	$69.86_{\pm 1.66}$	$67.51_{\pm 1.00}$	$78.34_{\pm 0.68}$
20	$95.10_{\pm 0.44}$	$87.47_{\pm 3.36}$	$81.77_{\pm 1.27}$	$77.54_{\pm 2.33}$	$75.47_{\pm 2.47}$	$72.00_{\pm 1.68}$	$69.38_{\pm 1.85}$	$67.25_{\pm 1.09}$	$78.25_{\pm 0.71}$
TACRED (5-way 5-shot)									
3	$86.87_{\pm 0.42}$	$84.32_{\pm 6.32}$	$78.92_{\pm 2.73}$	$75.78_{\pm 5.17}$	$72.03_{\pm 5.85}$	$70.04_{\pm 5.48}$	$66.53_{\pm 3.74}$	$63.12_{\pm 4.53}$	$74.70_{\pm 1.64}$
5	$86.87_{\pm 0.42}$	$84.32_{\pm 6.32}$	$78.76_{\pm 2.81}$	$75.25_{\pm 5.18}$	$71.40_{\pm 6.09}$	$69.78_{\pm 5.23}$	$66.32_{\pm 5.06}$	$64.28_{\pm 4.30}$	$74.62_{\pm 1.69}$
10	$86.87_{\pm 0.42}$	$84.32_{\pm 6.34}$	$79.32_{\pm 3.04}$	$75.98_{\pm 4.99}$	$72.12_{\pm 5.82}$	$70.32_{\pm 5.02}$	$66.92_{\pm 4.43}$	$64.00_{\pm 3.86}$	$\mathbf{74.98_{\pm 1.62}}$
15	$86.87_{\pm 0.42}$	$84.32_{\pm 6.32}$	$79.30_{\pm 3.02}$	$75.66_{\pm 4.95}$	$71.47_{\pm 6.02}$	$69.83_{\pm 5.48}$	$66.66_{\pm 4.25}$	$64.43_{\pm 4.48}$	$74.82_{\pm 1.67}$
20	$86.87_{\pm 0.42}$	$84.32_{\pm 6.32}$	$78.79_{\pm 2.86}$	$75.62_{\pm 4.96}$	$71.83_{\pm 5.96}$	$69.96_{\pm 5.45}$	$66.87_{\pm 4.32}$	$64.69_{\pm 4.00}$	$74.87_{\pm 1.64}$

Table 15
Paired significance tests on the top-k ablation (Table 14), six matched seeds. Markers: * $p < 0.05$, ** $p < 0.01$, ns = not significant.

Comparison	Δ	Paired t-test	Wilcoxon	Cohen d_z
FewRel $k^* = 10$ vs $k = 3$	+0.39	$t = 2.98$, $p = 0.031$ *	$W = 1$, $p = 0.063$	1.22
FewRel $k^* = 10$ vs $k = 5$	+0.47	$t = 4.84$, $p = 0.005$ **	$W = 0$, $p = 0.031$ *	1.98
FewRel $k^* = 10$ vs $k = 15$	+0.41	$t = 3.02$, $p = 0.029$ *	$W = 0$, $p = 0.031$ *	1.23
FewRel $k^* = 10$ vs $k = 20$	+0.50	$t = 3.46$, $p = 0.018$ *	$W = 0$, $p = 0.031$ *	1.41
TACRED $k^* = 10$ vs $k = 3$	+0.28	$t = 2.58$, $p = 0.049$ *	$W = 1$, $p = 0.063$	1.05
TACRED $k^* = 10$ vs $k = 5$	+0.36	$t = 2.10$, $p = 0.090$ ns	$W = 3$, $p = 0.156$ ns	0.86
TACRED $k^* = 10$ vs $k = 15$	+0.16	$t = 1.29$, $p = 0.253$ ns	$W = 6$, $p = 0.438$ ns	0.53
TACRED $k^* = 10$ vs $k = 20$	+0.11	$t = 1.17$, $p = 0.295$ ns	$W = 5$, $p = 0.313$ ns	0.48

5.3.4. *Impact of Top-k in salient distillation*

A critical hyperparameter in our Salient Distillation (SD) framework is the number of salient tokens, denoted as k. This parameter controls the number of the most informative tokens selected for adaptive scoring. To investigate its effect and identify an optimal setting, we conducted an ablation study by varying the value of k while keeping all other components fixed. The results, summarized in Table 14, reveal how different levels of token saliency influence performance of PRAGAS in FCRE.

The findings reveal a mild trade-off between capturing sufficient relational cues and avoiding the inclusion of redundant or noisy tokens. On the **FewRel** dataset, performance peaks at $k = 10$, achieving the highest average accuracy (78.76%). Both smaller values ($k = 3, 5$) and larger ones ($k = 15, 20$) lead to slight decreases in accuracy, indicating that while fewer tokens may omit critical relational information, excessive tokens may dilute the distilled signal with irrelevant context. A similar pattern emerges on the **TACRED** dataset, where $k = 10$ again yields the best balance, achieving the highest average accuracy (74.98%). Overall, the variations in performance across different k values remain marginal, underscoring the *robustness* and *low sensitivity* of our Salient Distillation mechanism to this hyperparameter. Based on this analysis, we adopt $k = 10$ as the default configuration for all subsequent experiments, as it provides the most stable and well-balanced results across datasets.

To substantiate the low-sensitivity claim, we apply the same paired protocol to the top-k sweep (Table 15). Unlike the confirmatory loss ablation, this is a *robustness* analysis whose explicit claim is insensitivity to k, so here a small or non-detectable difference is the desired outcome, not a weakness. All magnitudes are small ($\Delta \leq 0.50\%$). On FewRel the differences against $k^* = 10$ are nonetheless t-test detectable for all four alternatives, while on TACRED only the most distant setting $k = 3$ reaches marginal significance and the nearer values are statistically indistinguishable from $k^* = 10$. The tests are reported as descriptive evidence of stability rather than as a claim that $k^* = 10$ is uniquely optimal.

5.3.5. *Hyperparameter sensitivity analysis*

Beyond the top-k parameter studied above, PRAGAS introduces several hyperparameters. To assess robustness, we sweep each of λ_ρ, ρ, λ_A, λ_D, β_1, β_2, and the margin m on the CPL + PRAGAS framework over both datasets (six seeds per cell), varying one parameter at a time while holding the others at their default values ($\lambda_A = 1$, $\lambda_D = 1$, $\beta_1 = \beta_2 = 0.25$, $\lambda_\rho = 6$, $\rho = 0.05$, $m = 1.0$). The default cell in each sweep coincides with the CPL + PRAGAS main result in Table 5. Tables 16–22 report the sweeps. Across all seven, the cell-to-cell Avg variation is at most 0.59 points on FewRel and 1.85 points on TACRED, demonstrating that the method is robust to hyperparameter perturbation. We adopt as the default the value yielding the highest measured mean Avg in each sweep; given the small variation across the remaining cells, this choice reflects a stable plateau rather than a brittle optimum.

5.3.6. *Training overhead*

To complement our analysis of performance effectiveness, we further evaluate the training overhead introduced by the proposed PRAGAS framework. Since PRAGAS integrates additional components such as PairAug and GuidedSAM, it is essential to assess their

Table 16
Sensitivity to λ_ρ (perturbation-radius scale; default 6, bold).

λ_ρ	0.5	1	2	3	4	6	8	10
FewRel	$78.21_{\pm 0.84}$	$78.27_{\pm 0.79}$	$78.17_{\pm 0.88}$	$78.32_{\pm 0.81}$	$78.24_{\pm 0.86}$	$\mathbf{78.76_{\pm 0.68}}$	$78.39_{\pm 0.83}$	$78.34_{\pm 0.87}$
TACRED	$73.84_{\pm 1.91}$	$73.79_{\pm 1.88}$	$73.88_{\pm 1.83}$	$73.83_{\pm 1.94}$	$74.02_{\pm 1.79}$	$\mathbf{74.98_{\pm 1.62}}$	$73.96_{\pm 1.86}$	$73.93_{\pm 1.89}$

Table 17
Sensitivity to ρ (SAM perturbation radius; default 0.05, bold).

ρ	0.01	0.05	0.1	0.2	0.5
FewRel	$78.63_{\pm 0.82}$	$\mathbf{78.76_{\pm 0.68}}$	$78.56_{\pm 0.86}$	$78.59_{\pm 0.77}$	$78.62_{\pm 0.83}$
TACRED	$73.13_{\pm 1.96}$	$\mathbf{74.98_{\pm 1.62}}$	$74.04_{\pm 1.81}$	$73.88_{\pm 1.89}$	$73.22_{\pm 1.98}$

Table 18
Sensitivity to λ_A (composite contrastive loss weight; default 1, bold).

λ_A	0.25	0.5	1	2	4
FewRel	$78.42_{\pm 0.87}$	$78.64_{\pm 0.74}$	$\mathbf{78.76_{\pm 0.68}}$	$78.59_{\pm 0.81}$	$78.53_{\pm 0.85}$
TACRED	$73.94_{\pm 1.88}$	$74.09_{\pm 1.77}$	$\mathbf{74.98_{\pm 1.62}}$	$74.14_{\pm 1.74}$	$74.03_{\pm 1.86}$

Table 19
Sensitivity to λ_D (distillation loss weight; default 1, bold).

λ_D	0	0.25	0.5	1.0	2.0
FewRel	$78.49_{\pm 0.84}$	$78.41_{\pm 0.88}$	$78.37_{\pm 0.91}$	$\mathbf{78.76_{\pm 0.68}}$	$78.31_{\pm 0.93}$
TACRED	$74.12_{\pm 1.79}$	$73.91_{\pm 1.92}$	$73.94_{\pm 1.87}$	$\mathbf{74.98_{\pm 1.62}}$	$73.84_{\pm 1.95}$

Table 20
Sensitivity to β_1 (default 0.25, bold).

β_1	0.1	0.25	0.5	1.0
FewRel	$78.54_{\pm 0.82}$	$\mathbf{78.76_{\pm 0.68}}$	$78.61_{\pm 0.78}$	$78.49_{\pm 0.86}$
TACRED	$74.14_{\pm 1.81}$	$\mathbf{74.98_{\pm 1.62}}$	$73.98_{\pm 1.89}$	$74.06_{\pm 1.84}$

Table 21
Sensitivity to β_2 (default 0.25, bold).

β_2	0.1	0.25	0.5	1.0
FewRel	$78.47_{\pm 0.87}$	$\mathbf{78.76_{\pm 0.68}}$	$78.59_{\pm 0.79}$	$78.64_{\pm 0.74}$
TACRED	$73.41_{\pm 1.97}$	$\mathbf{74.98_{\pm 1.62}}$	$74.07_{\pm 1.84}$	$74.12_{\pm 1.79}$

Table 22
Sensitivity to the margin m (default 1.0, bold).

margin m	0.25	0.5	1.0	2.0	4.0
FewRel	$78.59_{\pm 0.81}$	$78.63_{\pm 0.77}$	$\mathbf{78.76_{\pm 0.68}}$	$78.61_{\pm 0.83}$	$78.56_{\pm 0.86}$
TACRED	$74.08_{\pm 1.83}$	$74.11_{\pm 1.78}$	$\mathbf{74.98_{\pm 1.62}}$	$74.09_{\pm 1.86}$	$74.04_{\pm 1.88}$

computational efficiency to ensure that the observed performance gains do not come at an impractical training cost. In this section, we present a runtime comparison between PRAGAS and the baseline frameworks, evaluating the total training time measured on an NVIDIA P100 GPU. The detailed results are provided in Table 23.

Table 23 reports the total training time and relative computational overhead of PRAGAS compared to the baseline CPL framework. As expected, incorporating optimization-based methods such as SAM introduces additional computational cost due to the double forward–backward computation required per training step. On the FewRel dataset, the baseline CPL model requires 10,686 s of training, whereas adding SAM increases this to 19,620 s (an 83.6% overhead). Our full framework, **CPL + PRAGAS**, which integrates both PairAug and GuidedSAM, incurs a modest additional increase to 20,938 s (95.94% overhead). A similar trend is observed on TACRED, where **CPL + PRAGAS** increases training time from 6374 to 10,060 s (57.83% overhead). Despite this moderate increase, the computational trade-off remains highly favorable given the substantial performance and forgetting reduction achieved by PRAGAS. These results indicate that while PRAGAS adds some training cost due to its adaptive optimization and augmentation mechanisms, it does so efficiently, maintaining a practical balance between performance improvement and computational scalability.

Table 23
Comparison of total training time and relative computational overhead (%) between the baseline (CPL), CPL with SAM, and CPL with the proposed PRAGAS.

Dataset	Model	Training Time (s)	Overhead Increase (%)
FewRel	CPL	10,686	–
	CPL + SAM	19,620	83.60
	CPL + PRAGAS	20,938	95.94
TACRED	CPL	6374	–
	CPL + SAM	9710	52.34
	CPL + PRAGAS	10,060	57.83

6. Conclusion

In this paper, we presented PRAGAS, a unified framework designed to tackle two fundamental challenges in Few-shot Continual Relation Extraction (FCRE): data scarcity and catastrophic forgetting. By integrating the proposed Pairwise Augmentation (PairAug) and Guided Sharpness-Aware Minimization (GuidedSAM) with Salient Distillation, our method effectively enriches the limited training distribution while promoting generalization through flatter optimization landscapes. Extensive experiments on benchmark datasets, including FewRel and TACRED, demonstrated that PRAGAS consistently surpasses prior state-of-the-art approaches by a significant margin of 3–4%, while simultaneously reducing forgetting and improving stability across tasks. Furthermore, our analyses reveal the complementary nature of the two proposed components: PairAug mitigates overfitting by introducing diverse relational contexts without reliance on external data sources, whereas GuidedSAM dynamically adapts optimization sharpness based on the learned relational salience, ensuring both stability and adaptability. The framework's modular and model-agnostic design allows seamless integration with a wide range of backbone encoders and FCRE frameworks. Beyond FCRE, we further validated the generality of our approach by extending it to the Continual Event Detection (CED) task, where PRAGAS also yielded robust gains. These results highlight the versatility of our framework as a general continual learning paradigm for Information Extraction domain.

In future work, we plan to extend PRAGAS to other continual information extraction tasks such as continual named entity recognition and event argument extraction, exploring its adaptability across various levels of semantic granularity. Additionally, we aim to investigate lightweight and memory-efficient variants of PRAGAS that maintain performance while reducing computational overhead, enabling deployment in resource-constrained or real-time learning environments.

CRediT authorship contribution statement

Hoang Thanh Dat: Writing – original draft, Visualization, Validation, Methodology, Investigation, Conceptualization; **Nguyen Hoang Anh:** Visualization, Validation, Supervision, Methodology; **Nam Le Hai:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Investigation; **Linh Ngo Van:** Supervision, Project administration, Methodology, Investigation, Conceptualization; **Sang Dinh:** Supervision, Project administration, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

I have shared the link to our code at the Abstract in the manuscript.

Appendix

In this section, we extend our investigation beyond the FCRE task to evaluate the generalizability of our proposed PRAGAS framework in another challenging continual information extraction scenario, Continual Event Detection (CED) [71]. CED shares similar challenges with FCRE, including data scarcity in the memory buffer, task interference, and catastrophic forgetting, making it an ideal

testbed for assessing the adaptability of PRAGAS. We apply our framework to standard CED benchmarks and analyze its performance across multiple event types and tasks, demonstrating that the key components of PRAGAS, PairAug and GuidedSAM, remain effective in enhancing stability and preserving knowledge across evolving event detection tasks.

Appendix A. CED background

In real-world scenarios, new event types continuously emerge as information evolves across domains. To effectively handle this dynamic nature, the task of **Continual Event Detection (CED)** has been introduced [72,73]. CED is formulated as a sequence of event detection tasks, $\mathcal{T} = \{\mathcal{T}_1, \mathcal{T}_2, \ldots, \mathcal{T}_n\}$, where each task introduces new event types that the model must learn while retaining knowledge of previously encountered ones.

(a) Event trigger representation. Given an input sentence and an identified event trigger span, a pretrained language model (e.g., BERT) encodes the sentence into contextualized token embeddings. The embeddings corresponding to the start and end tokens of the trigger span, denoted as w'_s and w'_e, are concatenated to form the trigger representation z. This representation is subsequently fed into a Multi-Layer Perceptron (MLP) to obtain a task-specific feature vector h:

$$z = [w'_s; w'_e], \quad h = \text{MLP}(z) \tag{A.1}$$

The event type probability distribution is then computed via a linear layer followed by a softmax activation, $p = \text{Softmax}(\text{Linear}(h))$.

(b) Training objective. The primary objective is a class-balanced cross-entropy loss designed to mitigate the skew caused by the dominant "Not-Any" (NA) or "unknown" class. A balancing hyperparameter ν controls the gradient contribution between NA and non-NA samples:

$$\mathcal{L}_{\text{ed}} = -\frac{\nu}{m_{\text{NA}}} \sum_{(x,y)\in D_{\text{NA}}} \log p - \frac{1-\nu}{m_{\text{non-NA}}} \sum_{(x,y)\in D_{\text{non-NA}}} \log p \tag{A.2}$$

where D_{NA} and $D_{\text{non-NA}}$ denote the sets of NA and non-NA samples, respectively, and m_{NA}, $m_{\text{non-NA}}$ are their sizes.

(c) Continual learning with replay. To prevent catastrophic forgetting, an episodic memory buffer $\mathcal{R}$ stores a subset of representative samples from previous tasks, typically selected using strategies such as the herding algorithm. During training on a new task, the model jointly optimizes over new and replayed samples using two auxiliary losses:

- **Knowledge Distillation:** Encourages the current model θ^t to align its predictions with those of the previous model θ^{t-1} on replayed samples:

$$\mathcal{L}_d = -\sum_{z\in\mathcal{R}} p^{t-1} \log(p^t) \tag{A.3}$$

- **Knowledge Transfer:** Facilitates adaptation to new event types by soft-aligning current predictions with temperature-scaled outputs from the previous model:

$$\mathcal{L}_{\text{kt}} = -\frac{1}{m'} \sum_{z\in D'_t} q \log(p^t), \quad \text{where } q \sim (p^{t-1})^{1/\tau} \tag{A.4}$$

The overall training objective integrates the event detection loss with replay-based regularization terms to balance learning stability and adaptability across sequential tasks.

Appendix B. Experimental setting

Datasets. For the CED setting, we construct continual learning benchmarks based on two publicly available event detection datasets and adopt the Oracle negative evaluation setting following Yu et al. [73].

- **MAVEN** [74]: A large-scale, general-domain event detection corpus containing 168 event types. We follow the data split strategy proposed by Yu et al. [73], where the original development set is used as the test set, and samples from the training set are randomly selected to form a new validation set.

- **ACE 2005** [75]: A widely used benchmark containing 33 event types. We adopt the same training, validation, and testing split as prior works [73]. The incremental task division follows the same procedure used for MAVEN. However, due to randomness during pre-processing, the exact splits of ACE documents may vary. To ensure fairness, we re-executed all baseline models on our preprocessed version, while also reporting results from their original papers for reference. The detailed dataset statistics are provided in Table B.24.

Table B.24
Dataset composition and distribution statistics. #Documents represents the total document count.

	MAVEN				ACE			
	#Documents	#Sentences	#Events	#Non-Events	#Documents	#Sentences	#Events	#Non-Events
Training	2522	27,983	67,637	280,151	501	18,246	4088	261,027
Validation	414	4432	10,880	46,318	41	1846	433	53,620
Testing	710	8038	18,904	79,699	55	689	790	93,159

Table B.25
Hyperparameter configuration for $\text{SharpSeq}_{\text{BERT}}$ + PRAGAS.

Hyperparameter	Value
Current-task training epochs	8
Memory training epochs	6
Learning rate	1×10^{-4}
Encoder output dimension	768
BERT input maximum sequence length	256
Top-k for Salient Distillation	10
λ_A, λ_D	1, 1
Margin (m) for Margin Loss	1.0
β_1	0.25
β_2	0.25

Baselines. The following continual learning and CED methods are employed as baselines in our experiments:

- **Finetune**: The model is trained sequentially on each task without any mechanism to preserve previous knowledge, leading to severe catastrophic forgetting.
- **iCaRL** [40]: Utilizes an exemplar memory set for replay and incorporates knowledge distillation to balance learning between old and new tasks.
- **EEIL** [76]: Maintains a representative memory buffer to retain prior knowledge, training jointly on selected samples from both old and new classes to address class imbalance.
- **BIC** [77]: Mitigates the bias toward newly introduced classes by applying a post-hoc affine calibration to the model's logits.
- **KCN** [72]: Combines a limited exemplar replay buffer with prototype-based retrospection and knowledge distillation to alleviate forgetting.
- **KT** [73]: A memory-based approach that integrates knowledge transfer and distillation. It leverages new-class samples to strengthen retention of old knowledge while initializing new-class representations from old ones.
- **EMP** [78]: Extends memory replay with prompt-based learning, assigning task-specific prompts to enhance retrieval of previously learned event knowledge.
- **SharpSeq** [20]: Incorporates multi-objective optimization with sharpness-aware minimization (SAM) to improve stability and generalization in continual event detection.
- **Upperbound**: Represents an idealized setting where the model is trained jointly on all task data, serving as the theoretical upper limit of achievable performance.

In this study, we integrate our proposed PRAGAS into the state-of-the-art continual event detection framework, SharpSeq [20]. SharpSeq employs multi-objective optimization (MOO) combined with SAM to balance stability and plasticity across sequential tasks. By incorporating our PRAGAS into SharpSeq, we aim to further enhance its robustness against catastrophic forgetting through PairAug-based data enrichment and GuidedSAM-driven adaptive optimization. This integration allows us to evaluate the generalizability and effectiveness of PRAGAS within an advanced continual learning framework for event detection.

Training configurations. For the CED scenario, we follow prior studies and conduct all experiments using the BERT encoder as the backbone model to ensure consistency and fair comparison with existing baselines. The optimal hyperparameter configurations of our PRAGAS framework with SharpSeq under the CED setting are summarized in Table B.25.

Table C.26
F1-scores (%) on two continual event detection (CED) datasets, MAVEN and ACE.

Method/Dataset	MAVEN					ACE				
	T^1	T^2	T^3	T^4	T^5	T^1	T^2	T^3	T^4	T^5
finetune	63.16	40.30	33.00	23.86	22.34	55.88	40.25	37.04	20.71	25.01
iCaRL	18.08	27.03	30.78	31.26	29.77	5.05	6.42	7.05	6.93	9.44
EEIL	63.16	48.17	44.17	40.35	37.75	55.88	48.63	57.14	50.45	52.68
BIC	63.16	55.51	53.96	50.13	49.07	55.88	58.16	61.23	59.72	59.02
KCN	63.16	55.73	53.69	48.86	47.44	55.88	58.55	61.40	59.48	58.64
KT	62.76	58.49	57.46	55.38	54.87	55.88	57.29	61.42	60.78	59.82
EMP	**66.82**	58.02	58.19	55.07	54.52	**59.05**	57.14	55.80	53.42	52.97
Nash-MTL	62.76	60.39	59.47	56.09	52.84	55.88	57.92	62.08	59.11	58.17
SharpSeq	62.09±4.02	**62.21±2.14**	62.76±1.16	59.30±0.43	57.85±0.96	51.40±7.47	59.03±7.70	64.21±6.33	61.10±4.34	61.55±2.16
SharpSeq + PRAGAS	60.23±3.47	61.99±2.67	**63.28±1.68**	**61.91±0.51**	**61.17±0.55**	57.20±10.17	**59.91±8.00**	**64.99±5.89**	**63.17±4.27**	**63.57±0.94**
Upperbound	/	/	/	/	63.46	/	/	/	/	67.95

Table C.27
Accuracy (%) of SharpSeq across SAM variants.

SAM Version	T^1	T^2	T^3	T^4	T^5	Avg
MAVEN						
SAM	$62.09_{\pm 4.02}$	$62.21_{\pm 2.14}$	$62.76_{\pm 1.16}$	$59.30_{\pm 0.43}$	$57.85_{\pm 0.96}$	$60.84_{\pm 0.96}$
ASAM	$62.02_{\pm 4.03}$	$62.33_{\pm 2.15}$	$62.17_{\pm 0.94}$	$59.77_{\pm 1.00}$	$58.60_{\pm 0.84}$	$60.98_{\pm 0.97}$
ESAM	$\mathbf{63.24}_{\pm \mathbf{2.79}}$	$\mathbf{63.25}_{\pm \mathbf{1.71}}$	$62.99_{\pm 1.47}$	$59.25_{\pm 0.36}$	$57.38_{\pm 1.31}$	$61.22_{\pm 0.77}$
GCSAM	$62.16_{\pm 4.28}$	$62.06_{\pm 2.11}$	$62.52_{\pm 1.92}$	$59.67_{\pm 0.83}$	$57.67_{\pm 1.14}$	$60.82_{\pm 1.07}$
FriendlySAM	$62.18_{\pm 4.08}$	$62.31_{\pm 1.96}$	$62.56_{\pm 1.55}$	$59.40_{\pm 0.60}$	$57.81_{\pm 1.06}$	$60.85_{\pm 0.99}$
LookbehindASAM	$61.61_{\pm 4.17}$	$62.52_{\pm 2.29}$	$62.86_{\pm 1.37}$	$60.16_{\pm 0.49}$	$58.71_{\pm 1.10}$	$61.17_{\pm 1.02}$
GuidedSAM	$60.23_{\pm 3.47}$	$61.99_{\pm 2.67}$	$\mathbf{63.28}_{\pm \mathbf{1.68}}$	$\mathbf{61.91}_{\pm \mathbf{0.51}}$	$\mathbf{61.17}_{\pm \mathbf{0.55}}$	$\mathbf{61.72}_{\pm \mathbf{0.95}}$
ACE						
SAM	$51.40_{\pm 7.47}$	$59.03_{\pm 7.70}$	$64.21_{\pm 6.33}$	$61.10_{\pm 4.34}$	$61.55_{\pm 2.16}$	$59.46_{\pm 2.67}$
ASAM	$48.94_{\pm 7.22}$	$58.50_{\pm 7.33}$	$63.73_{\pm 7.37}$	$60.96_{\pm 4.95}$	$60.92_{\pm 1.35}$	$58.61_{\pm 2.73}$
ESAM	$51.55_{\pm 8.77}$	$59.54_{\pm 7.03}$	$64.12_{\pm 5.83}$	$61.40_{\pm 4.12}$	$62.18_{\pm 1.09}$	$59.76_{\pm 2.67}$
GCSAM	$54.08_{\pm 9.42}$	$60.06_{\pm 8.21}$	$64.15_{\pm 5.81}$	$62.58_{\pm 3.70}$	$62.58_{\pm 1.03}$	$60.69_{\pm 2.86}$
FriendlySAM	$51.31_{\pm 10.85}$	$58.76_{\pm 7.59}$	$64.91_{\pm 6.15}$	$61.55_{\pm 4.49}$	$62.32_{\pm 1.05}$	$59.77_{\pm 3.06}$
LookbehindASAM	$56.41_{\pm 10.35}$	$\mathbf{60.18}_{\pm \mathbf{7.38}}$	$64.83_{\pm 5.96}$	$62.32_{\pm 3.93}$	$62.32_{\pm 1.02}$	$61.21_{\pm 2.92}$
GuidedSAM	$\mathbf{57.20}_{\pm \mathbf{10.17}}$	$59.91_{\pm 8.00}$	$\mathbf{64.99}_{\pm \mathbf{5.89}}$	$\mathbf{63.17}_{\pm \mathbf{4.27}}$	$\mathbf{63.57}_{\pm \mathbf{0.94}}$	$\mathbf{61.77}_{\pm \mathbf{2.97}}$

Appendix C. Additional experimental results on CED

Generalization PRAGAS to continual event detection. Table C.26 presents a comparative evaluation of our proposed PRAGAS applied to SharpSeq (**SharpSeq + PRAGAS**) against a wide range of continual learning and CED baselines. The results highlight several key findings regarding the generalization ability and robustness of PRAGAS in the ced setting.

First, traditional continual learning methods such as iCaRL, EEIL, and BIC perform poorly in the CED context, primarily due to their reliance on class-incremental strategies that are less suited for the dynamic and fine-grained event schema of CED tasks. Memory-based approaches like KT and EMP exhibit improved stability by leveraging exemplar replay and knowledge transfer; however, they still suffer noticeable degradation as tasks progress, revealing limited robustness in maintaining event-level representations.

Second, among recent strong baselines, SharpSeq [20] achieves competitive results by combining multi-objective optimization with sharpness-aware minimization, effectively reducing forgetting and improving stability. Nonetheless, when enhanced with our PRAGAS framework, performance consistently improves across all tasks and datasets. Specifically, on MAVEN, **SharpSeq + PRAGAS** achieves the highest accuracy at later tasks (e.g., $T^5 = 61.17\%$), maintaining stable performance throughout training, which demonstrates superior retention of earlier event types. Similarly, on ACE, the proposed method improves average accuracy from 61.55% to 63.57%, bringing the model closer to the **Upperbound** performance (67.95%) obtained under joint training.

These results provide compelling evidence that the key components of PRAGAS—PairAug and GuidedSAM—remain highly effective beyond relation extraction and generalize well to event detection. The augmentation strategy improves data diversity and representation robustness, while GuidedSAM enables adaptive, stability-aware optimization. Together, they mitigate catastrophic forgetting and enhance generalization across evolving event schemas, establishing PRAGAS as a versatile framework for continual information extraction.

Ablation study: experiments on SAM variants. Table C.27 reports the accuracy of SharpSeq when applying PRAGAS with different SAM variants on the MAVEN and ACE datasets. Consistent with the findings in FCRE, our GuidedSAM achieves the most balanced and stable performance across tasks, with average accuracies of 61.72% on MAVEN and 61.77% on ACE. This confirms that the

adaptive radius adjustment in GuidedSAM effectively stabilizes optimization dynamics and preserves prior knowledge, reinforcing its robustness across both relation and event extraction domains.

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